

Graph Neural Networks for the Classification And Lead Identification of EGFR Inhibitors in Non-Small Cell Lung Cancer



**A Thesis Submitted for the Partial Fulfillment of the
Requirements for the Master of Science (M. Sc.)
Degree in Data Science**

**Submitted to the
Central Department of Mathematics /
Kirtipur Campus
Institute of Science And Technology
Tribhuvan University
NEPAL**

By

**Krishna Gaire
Symbol No.: 7916015
T.U. Registration No.: 5-3-28-49-2022**

AUGUST, 2026

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Declaration

This thesis entitled “**Graph Neural Networks for the Classification And Lead Identification of EGFR Inhibitors in Non-Small Cell Lung Cancer**” which is being submitted to the Department of Mathematics, Kirtipur Campus, Institute of Science and Technology (IoST), Tribhuvan University, Nepal for the partial fulfillment of the requirement to the Master of Science Degree in Data Science has been carried out by me under the supervision of **Pravat Upreti** at Central Department of Mathematics, Kirtipur Campus. This work is original and has not been submitted earlier in part or full in this or any other form to any other institution for the award of any degree.

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Supervisor's Recommendation

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Letter of Approval

The thesis work entitled “**Graph Neural Networks for the Classification And Lead Identification of EGFR Inhibitors in Non-Small Cell Lung Cancer**” submitted at the Institute of Science and Technology, Department of Mathematics, Tribhuvan University by **Krishna Gaire**, University Registration Number **5-3-28-49-2022**, has been accepted for the partial fulfillment of Master in Data Science.

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Abstract

Non-small cell lung cancer (NSCLC) accounts for about 85% of lung cancer cases, and EGFR mutations are among its most significant oncogenic drivers. EGFR-targeted tyrosine kinase inhibitors have transformed treatment, yet acquired resistance and high therapy cost remain major challenges, especially in resource-limited South Asian settings. This study presents a graph neural network (GNN) framework for EGFR-inhibitor classification and the prospective virtual screening of compounds from traditional Nepali medicinal plants.

A curated dataset of 10,523 compounds with validated EGFR bioactivity (74% active) was assembled from ChEMBL, standardised with RDKit, and represented as molecular graphs with 29-dimensional atom and 12-dimensional bond features. Seven models were benchmarked under a Bemis-Murcko scaffold split: five GNNs (GCN, GAT, GIN, AttentiveFP, GraphSAGE) and two ECFP4 baselines (Random Forest, MLP). The baselines led (Random Forest AUROC 0.8827, MLP 0.8547) ahead of the GNNs (0.8087–0.8454, GCN strongest); Optuna tuning raised AttentiveFP’s validation AUROC to 0.8632 but did not surpass them—well-tuned fingerprint models remain highly competitive at this dataset size.

The two best models (GCN and Random Forest) were applied to a natural-product library of seven Nepali plant species from COCONUT 2.0. Of 535 retrieved compounds, eight already in the ChEMBL training data—including the apparent top hits apigenin and kaempferol—were excluded as leakage, leaving 527 genuinely unseen compounds. The hit list was strongly model-dependent (Random Forest 179 predicted actives versus GCN 24), and neither best model gave a Lipinski-compliant lead at probability ≥ 0.80 . The strongest candidates are the xanthone **isogentisin** (Random Forest 0.728) and the flavone **luteolin**, the top cross-model consensus lead—both at moderate confidence.

Model-consensus screening can thus surface drug-like natural-product candidates from Nepali ethnomedicinal libraries using an open-source, CPU-only pipeline, while showing that such screens are model-dependent, must exclude training-set compounds to stay prospective, and yield only moderate-confidence leads requiring experimental validation.

Keywords: *EGFR inhibitors, graph neural networks, virtual screening, Nepali medicinal plants, drug discovery.*

शोधसार

फोक्सोको क्यान्सरमध्ये सबैभन्दा बढी पाइने नन-स्मल सेल लङ क्यान्सर (NSCLC) मा EGFR उत्पत्तिवर्तन प्रमुख कारक हो, तर अवरोधकहरूप्रतिको प्रतिरोध र उपचारको उच्च लागत ठूलो चुनौती बनेको छ। यस अध्ययनले EGFR अवरोधकको वर्गीकरण र नेपाली औषधीय वनस्पतिका यौगिकहरूको भर्चुअल स्क्रिनिङका लागि ग्राफ न्युरल नेटवर्क (GNN) ढाँचा प्रस्तुत गर्दछ। ChEMBL बाट लिइएका १०,५२३ यौगिकमा Bemis-Murcko स्क्वाफोल्ड विभाजन प्रयोग गरी पाँच GNN र दुई आधारभूत मोडेल गरी सात मोडेलको तुलना गरियो। फिंगरप्रिन्ट-आधारित Random Forest (AUROC 0.8827) ले सर्वोत्तम प्रदर्शन गर्दै सबै GNN लाई उछिन्यो। दुई उत्कृष्ट मोडेलद्वारा सात नेपाली वनस्पतिका ५२७ नयाँ यौगिकको स्क्रिनिङ गर्दा कुनै पनि यौगिकले $p \geq 0.80$ पार गरेन; xanthone isogentisin र flavone luteolin मध्यम-विश्वासका सम्भावित यौगिक रहे। यसले GNN-आधारित विधिले नेपाली परम्परागत वनस्पतिबाट औषधि-योग्य सम्भावित यौगिक पहिचानमा सहयोग गर्न सक्ने देखाउँछ।

मुख्य शब्दहरू: EGFR अवरोधक, ग्राफ न्युरल नेटवर्क, भर्चुअल स्क्रिनिङ, नेपाली औषधीय वनस्पति, औषधि खोज

List of Abbreviations and Acronyms

Abbreviation	Full Form
ADMET	Absorption, Distribution, Metabolism, Excretion, and Toxicity
ATP	Adenosine Triphosphate
AUPRC	Area Under the Precision–Recall Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
C797S	Cysteine-797 to Serine substitution (EGFR resistance mutation)
ChEMBL	Chemical European Molecular Biology Laboratory
COCONUT	Collection of Open Natural prodUcTs (natural product database)
CPU	Central Processing Unit
ECFP	Extended Connectivity Fingerprint
ECFP4	Extended Connectivity Fingerprint, radius 2 (4-bond diameter)
EGF	Epidermal Growth Factor
EGFR	Epidermal Growth Factor Receptor
ErbB	Erythroblastic Leukaemia Viral Oncogene Homologue (receptor family)
Ex19del	Exon 19 Deletion (EGFR sensitising mutation)
F1	F1 Score (harmonic mean of precision and recall)
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GIN	Graph Isomorphism Network
GNN	Graph Neural Network
GPU	Graphics Processing Unit
GraphSAGE	Graph SAmple and aggreGatE
GRU	Gated Recurrent Unit
HBA	Hydrogen Bond Acceptors
HBD	Hydrogen Bond Donors
HER2	Human Epidermal Growth Factor Receptor 2 (ErbB2)
IC ₅₀	Half-Maximal Inhibitory Concentration
IoST	Institute of Science and Technology
L858R	Leucine-858 to Arginine substitution (EGFR mutation)
MCC	Matthews Correlation Coefficient
ML	Machine Learning
MLP	Multi-Layer Perceptron
MW	Molecular Weight
NSCLC	Non-Small Cell Lung Cancer
pIC ₅₀	Negative Logarithm of IC ₅₀
QED	Quantitative Estimate of Drug-likeness

ReLU	Rectified Linear Unit
RF	Random Forest
Ro5	Lipinski's Rule of Five
SMILES	Simplified Molecular Input Line Entry System
T790M	Threonine-790 to Methionine substitution (EGFR gate-keeper mutation)
TKI	Tyrosine Kinase Inhibitor
TPSA	Topological Polar Surface Area
TU	Tribhuvan University
VS	Virtual Screening

List of Symbols

Symbol	Description
$\hat{p}$	Predicted probability of activity from the classifier
d_n	Dimensionality of atom node feature vectors (29 in this work)
d_e	Dimensionality of bond edge feature vectors (12 in this work)
n	Number of compounds
$\text{Log } P$	Logarithm of the octanol–water partition coefficient
$\text{\AA}$	Ångström (10^{-10} m)
$\text{\AA}^2$	Square Ångström (area unit for TPSA)
Da	Dalton (atomic mass unit)
nM	Nanomolar (10^{-9} mol/L)
μM	Micromolar (10^{-6} mol/L)

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Chapter 1

Introduction

1.1 Background

Cancer remains one of the most formidable public health challenges of the twenty-first century, imposing an extraordinary clinical, economic, and societal burden on populations worldwide. According to the GLOBOCAN 2022 estimates compiled by the International Agency for Research on Cancer (IARC), there were approximately 20 million new cancer cases and 9.7 million cancer-related deaths globally in 2022 (Siegel et al., 2023). Among all cancer subtypes, **lung cancer** is the leading cause of cancer mortality, accounting for approximately 1.8 million deaths and representing nearly 18% of all cancer fatalities in that year. It is the second most commonly diagnosed cancer by incidence and the most lethal, primarily because the majority of patients are diagnosed at advanced or metastatic stages when curative surgery is no longer feasible.

Non-Small Cell Lung Cancer (NSCLC) is the predominant histological category of lung cancer, constituting approximately 85% of all diagnoses. NSCLC encompasses three major subtypes: adenocarcinoma (~40%), squamous cell carcinoma (~25–30%), and large-cell carcinoma (~10–15%). The remaining 15% is classified as Small Cell Lung Cancer (SCLC), a neuroendocrine tumour with distinct biology and treatment approaches. The epidemiology of NSCLC has important geographical and ethnic dimensions: the disease burden is particularly acute across South and East Asia, including Nepal, where limited access to computed tomography (CT) screening programmes and low public awareness result in late-stage diagnosis in the majority of patients, precluding surgical intervention and significantly worsening prognosis (Rotow & Bivona, 2023). The five-year overall survival rate for stage IV NSCLC remains below 10% with conventional chemotherapy.

1.1.1 The Epidermal Growth Factor Receptor in NSCLC

The **Epidermal Growth Factor Receptor (EGFR)** is a transmembrane receptor tyrosine kinase (RTK). Upon ligand binding, EGFR undergoes dimerization and intrinsic kinase activation, which initiates key intracellular downstream pathways—specifically RAS–RAF–MEK–ERK, PI3K–AKT–mTOR, STAT3/5, and PLC- γ —that orchestrate cellular proliferation, motility, and survival (Rotow & Bivona, 2023). In non-small cell lung cancer (NSCLC), somatic activating mutations within the EGFR kinase domain induce constitutive, ligand-independent receptor activation, thereby acting as primary oncogenic drivers. Two major sensitizing mutations account for 85–90% of all EGFR-mutant NSCLC cases: **exon 19 deletions** (Ex19del; ~45–50%) and the **L858R** point mutation in the activation loop (~40%).

EGFR-mutation prevalence varies sharply by ethnicity—10–15% in Caucasian NSCLC versus 40–50% in East Asian populations (Leonetti et al., 2019). Given Nepal's rising NSCLC incidence and its admixed South/East Asian population, a large and growing

share of Nepali patients harbour targetable EGFR mutations, yet molecular diagnostics and novel targeted therapies remain scarce—making affordable computational drug discovery particularly relevant.

1.1.2 Generations of EGFR Tyrosine Kinase Inhibitors

EGFR TKIs have developed in three generations, each overcoming resistance to its predecessor. **First-generation** reversible inhibitors (gefitinib, erlotinib) achieved 55–75% response rates and 9–13-month progression-free survival in sensitising-mutant NSCLC, but resistance emerged in nearly all patients—most often (~50–60%) via the **T790M** gatekeeper mutation. **Second-generation** covalent inhibitors (afatinib, dacomitinib) bind Cys797 irreversibly but their wild-type EGFR inhibition causes dose-limiting toxicity. **Third-generation** covalent inhibitors, led by **osimertinib** (FDA-approved 2015), selectively target T790M-mutant over wild-type EGFR and, with superior efficacy (PFS 18.9 vs. 10.2 months) and CNS penetration, are now the standard of care (Rotow & Bivona, 2023).

The C797S resistance problem. Despite osimertinib’s superiority, acquired resistance emerges in virtually all patients within 9–18 months. The most common resistance mechanism is the **C797S substitution** at the covalent binding site (Cys797 → Ser797), which abrogates the irreversible covalent bond formation required for osimertinib activity. When C797S occurs in *cis* with T790M (on the same allele), the resulting **triple-mutant EGFR** forms (Ex19del/T790M/C797S and L858R/T790M/C797S) are *resistant to all three generations* of currently approved EGFR TKIs. Additional resistance mechanisms include MET amplification, HER2 amplification, RAS/RAF mutations, and SCLC histological transformation (Leonetti et al., 2019). As of 2026, there is no approved therapy for triple-mutant EGFR, and it remains a major long-term unmet need in NSCLC treatment. It is important, however, to state the scope of this thesis precisely at the outset: because no large-scale bioactivity dataset yet exists for the triple mutant, the present work develops and validates its computational framework on the abundant *aggregate* EGFR inhibition data available in ChEMBL (ChEMBL203)—which is dominated by wild-type and single-mutant measurements. The framework therefore targets *general* EGFR inhibitor identification; extending it to mutant-specific prediction, including the triple mutant, is a direction for future work rather than a claim of this study.

1.1.3 Computational Drug Discovery and Graph Neural Networks

Traditional drug discovery is a slow, costly, and attrition-prone process. The average time from target identification to regulatory approval is 10–15 years, and the average cost per approved drug has been estimated at approximately USD 2.0–2.6 billion (Schneider & Clark, 2020). The high attrition rate (~90% of compounds entering Phase I clinical trials fail) is driven primarily by inadequate efficacy, unexpected toxicity, and poor pharmacokinetic properties, all of which are increasingly addressable through computational prediction.

Computer-Aided Drug Design (CADD) encompasses a broad spectrum of computational techniques including molecular docking, molecular dynamics simulation, free energy perturbation, QSAR modelling, and virtual screening. Among these, machine

learning-based virtual screening has emerged as the most scalable approach for the rapid prioritisation of large compound libraries (Walters & Murcko, 2020).

Graph Neural Networks (GNNs) represent a particularly powerful class of machine learning architectures for molecular property prediction (Gilmer et al., 2017). The fundamental insight is that molecules can be represented as mathematical graphs in which atoms correspond to nodes carrying feature vectors and chemical bonds correspond to edges carrying their own feature vectors, as illustrated in Figure 1. GNNs learn molecular representations through iterative message-passing: at each layer, every atom aggregates feature information from its bonded neighbours and updates its own hidden representation accordingly. After several such message-passing layers, a readout step aggregates all the atom representations into a single fixed-length molecular embedding that is passed to a classifier or regression head.

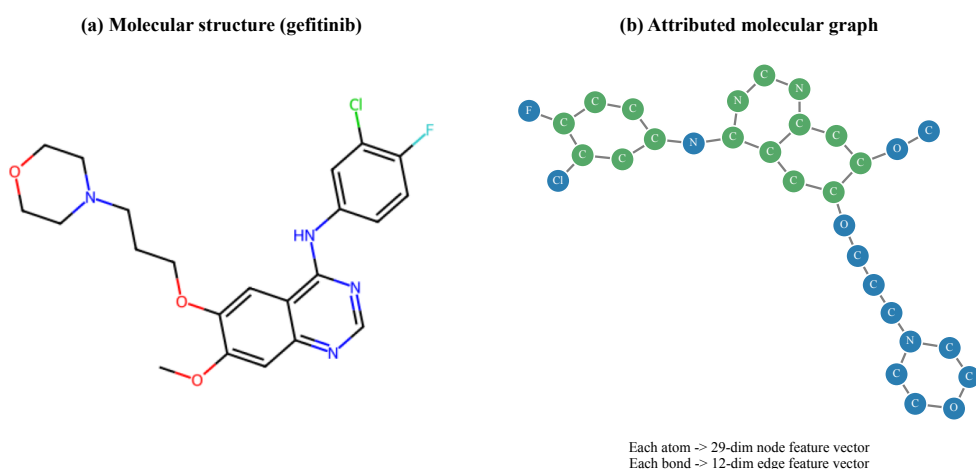


Figure 1: A molecule (the EGFR inhibitor gefitinib) represented as an attributed graph of atom nodes and bond edges.

This approach offers several advantages over traditional fingerprint-based molecular representations. First, GNNs learn task-specific molecular representations *end-to-end* from bioactivity data, rather than relying on pre-defined structural patterns encoded in fingerprints. Second, by incorporating edge feature vectors, GNNs can natively encode bond-level information (bond order, conjugation, stereochemistry) that is either discarded or only coarsely approximated in circular fingerprints. Third, deep GNN architectures can capture long-range atomic interactions spanning multiple bonds, enabling them to model higher-order pharmacophoric relationships that influence molecular activity (Jiang et al., 2021).

Despite these advantages, the translation of GNNs from academic benchmarks to practical drug discovery applications faces challenges, including the need for rigorous evaluation protocols, chemical interpretability, and extension to prospective screening pipelines—all of which are directly addressed by this thesis.

1.1.4 Nepali Medicinal Plants as a Natural-Product Resource

Nepal’s Himalayan biodiversity spans roughly 6,500 flowering plant species, of which more than 700 have documented ethnopharmacological use in Ayurvedic and Tibetan

medicine. Many are inexpensive, culturally familiar, and locally accessible, making them an attractive yet under-explored source of natural-product leads for resource-limited settings. The prospective screening dataset used in this thesis is drawn from **seven traditional Nepali medicinal plant species**—*Swertia chirayita*, *Rhododendron arboreum*, *Nardostachys jatamansi*, *Terminalia chebula*, *Berberis aristata*, *Ocimum sanctum*, and *Tinospora cordifolia*—selected for their documented anti-inflammatory and antiproliferative use and their coverage in the **COCONUT 2.0** open natural-product database (Sorokina et al., 2021). After RDKit standardisation this yields a library of **535 unique natural-product compounds** (ranging from 9 compounds for *Ocimum sanctum* to 203 for *Nardostachys jatamansi*) that is screened for EGFR inhibitor activity in Chapter 4. Screening this locally accessible chemical space directly connects computational drug discovery to Nepal’s biodiversity and offers an affordable route to candidate anti-NSCLC compounds.

1.2 Problem Statement

Despite the transformative impact of EGFR-targeted therapy and the promise of Graph Neural Networks for drug discovery, critical and interrelated gaps persist in both clinical practice and computational methodology.

1.2.1 Clinical and Translational Need

Acquired resistance to EGFR tyrosine kinase inhibitors—most notably the **C797S mutation**, which renders third-generation covalent inhibitors ineffective and for which no fourth-generation drug is yet approved—sustains a continuing need to discover structurally novel, non-covalent EGFR inhibitor chemotypes (Rotow & Bivona, 2023). This need is most acute in South and East Asia, where EGFR-mutant NSCLC is highly prevalent yet novel targeted therapies remain expensive and computational drug-discovery infrastructure is scarce. There is therefore a clear demand for open-source, CPU-executable screening pipelines that can prioritise affordable, regionally accessible natural-product leads—the translational problem this thesis addresses.

1.2.2 Computational Problem: Methodological Gaps in GNN-Based EGFR Inhibitor Prediction

A review of the existing computational literature reveals three specific methodological deficiencies that this thesis is designed to address:

1. **Evaluation bias from random splitting.** The majority of published GNN models for molecular property prediction, including those applied to EGFR inhibitor classification, employ random train/test splitting. This approach allows structurally similar or near-identical molecular scaffolds to appear simultaneously in the training and test sets. As a result, the reported performance metrics substantially overestimate a model’s ability to generalise to genuinely novel chemical matter—precisely the task required in prospective virtual screening. A model that has “memorised” a scaffold from the training set cannot be trusted to correctly classify a structurally unprecedented compound in a real screening campaign.

2. **Limited molecular featurisation.** Many published GNN models for EGFR or related kinase targets use minimal node feature sets (e.g., atomic number only, or a small set of atom type flags) and ignore edge (bond) features entirely. This discards potentially discriminative chemical information including bond order, conjugation, aromaticity, and stereochemistry—all of which influence molecular recognition in the EGFR binding pocket.
3. **Absence of prospective virtual screening on natural product libraries.** Most academic GNN studies in the EGFR domain report classification performance on held-out test sets but do not extend their trained models to the most practically relevant application: scoring and ranking an external library of unseen compounds to identify novel lead candidates—especially compounds derived from traditional medicinal plants that may offer affordable, regionally accessible therapeutic options (Malik et al., 2025).

This thesis addresses these gaps through its two objectives. First, it *builds graph neural network classifiers*—with rich 29-dimensional atom and 12-dimensional bond featurisation, evaluated under a strict Bemis-Murcko scaffold split—and *benchmarks them against traditional machine-learning baselines* (Random Forest and an MLP on ECFP4 fingerprints), directly testing whether the added complexity of GNNs is justified over simpler, well-established models (gaps 1–2). Second, it *deploys the best-performing models to screen* a Nepali medicinal-plant compound library for drug-like EGFR inhibitor leads (gap 3). Because large-scale triple-mutant bioactivity data are not yet available, the models are trained on aggregate EGFR data, and the natural-product scaffolds they surface are intended as candidate starting points for future experimental validation rather than as validated triple-mutant inhibitors.

1.3 Objectives

1.3.1 General Objective

To build and benchmark a Graph Neural Network framework for EGFR inhibitor classification against traditional machine-learning baselines, and to deploy the best-performing models for prospective virtual screening of Nepali medicinal plant compounds as candidate anti-NSCLC leads.

1.3.2 Specific Objectives

1. To build graph neural network models that classify EGFR inhibitors and compare them with traditional machine-learning models.
2. To use the best models to screen Nepali medicinal plant compounds and identify drug-like EGFR inhibitor leads.

1.4 Rationale

Clinical rationale. Acquired resistance to EGFR TKIs—and ultimately the triple-mutant EGFR that follows osimertinib failure—continues to drive the search for

structurally novel inhibitor chemotypes, yet bringing new therapies through conventional drug discovery takes years to decades. Computational virtual screening can accelerate the early stage of this search by pre-selecting the most promising candidates for experimental testing, reducing the number of compounds that must be synthesised and assayed. In Nepal and South Asia, where EGFR mutations are prevalent yet targeted therapies remain expensive, this further motivates screening locally accessible, low-cost natural-product chemical space rather than costly synthetic libraries (Rotow & Bivona, 2023).

Scientific rationale. Graph Neural Networks have demonstrated strong performance across molecular machine learning benchmarks (Hu et al., 2020), yet their application to EGFR inhibitor discovery has been constrained by the three methodological shortcomings enumerated in Section 1.2. Addressing all three simultaneously within a single, reproducible pipeline—scaffold splitting, rich featurisation, systematic multi-architecture comparison, and prospective Nepali plant compound screening—is what motivates the design of this study.

Data science rationale. This project integrates multiple core data science competencies within a single applied research context: large-scale database querying and API integration (ChEMBL), cheminformatics feature engineering (RDKit), deep learning architecture design and training (PyTorch, PyTorch Geometric), automated hyperparameter search (Optuna), prospective natural compound screening (COCONUT 2.0), and scientific visualisation (Matplotlib, Seaborn, RDKit 2D depictions). The project therefore constitutes a comprehensive demonstration of advanced data science methodology applied to a high-impact biomedical problem.

Practical rationale. The pipeline was deliberately built from exclusively open-source Python libraries (RDKit, PyTorch Geometric, Optuna, scikit-learn) and public databases (ChEMBL, COCONUT 2.0), running on commodity CPU hardware, so that every reported result can be reproduced without licensed software or specialised computing resources.

1.5 Research Hypotheses

The following two testable research hypotheses were formulated prior to experimentation and are evaluated against the results presented in Chapter 4. Each is stated as a null hypothesis (H_0) and an alternative hypothesis (H_1).

1. H_1 (Performance Hypothesis).

- H_0 : the best graph neural network does *not* achieve a higher scaffold test-set AUROC than the best fingerprint baseline (Random Forest or MLP).
- H_1 : the best graph neural network achieves a higher scaffold test-set AUROC than the best fingerprint baseline.

2. H_2 (Virtual Screening Hypothesis).

- H_0 : no genuinely unseen, Lipinski-compliant Nepali plant compound is predicted active (predicted probability $\hat{p} \geq 0.80$ and QED ≥ 0.40) by *both* best models.

- H_1 : at least one genuinely unseen, Lipinski-compliant Nepali plant compound is predicted active ($\hat{p} \geq 0.80$ and QED ≥ 0.40) by *both* best models, providing a candidate lead for affordable NSCLC therapy.

1.6 Significance of the Study

This research makes several original and complementary contributions to the intersection of computational data science and molecular drug discovery:

Methodological contribution. Although systematic comparisons of GNN architectures on molecular benchmarks are well established (Jiang et al., 2021), this thesis applies such a comparison, to the author’s knowledge for the first time, *specifically to EGFR inhibitor classification*: it evaluates five GNN architectures (GCN, GAT, GIN, AttentiveFP, GraphSAGE) against traditional fingerprint baselines (Random Forest and MLP) under a rigorous Bemis-Murcko scaffold split, using a rich 29-dimensional atom and 12-dimensional bond featurisation and Bayesian hyperparameter optimisation (Optuna) for reproducible model selection. The methodological novelty lies in this EGFR-specific, scaffold-aware, baseline-anchored evaluation—coupled directly to the prospective natural-product screen below—rather than in the architecture comparison in isolation.

Natural product screening contribution. By curating bioactive compounds from seven traditional Nepali medicinal plant species (COCONUT 2.0) and applying the study’s best-performing models (the best graph network and the best overall classifier) for prospective EGFR inhibition prediction, this thesis delivers the first computational evaluation of Nepali natural products as potential EGFR TKI candidates—directly relevant to the South Asian oncology community where EGFR mutations are prevalent and novel therapies remain expensive.

Translational contribution. By extending the best-performing models into a complete multi-model prospective virtual screening pipeline applied to Nepali medicinal plant compounds and generating a ranked list of Lipinski-compliant natural-product-derived lead candidates with computed ADMET-proxy properties, the thesis delivers a concrete, actionable output—a prioritised shortlist of natural compounds for future experimental validation—rather than a purely academic performance comparison.

Regional and social contribution. The elevated prevalence of EGFR-activating mutations in East and South Asian NSCLC patients, combined with the rising cancer burden and limited access to novel therapies in Nepal, makes this research particularly relevant to the South Asian academic and clinical community. The entirely open-source, CPU-executable implementation ensures that the methodology can be adopted and extended by researchers in resource-limited settings without access to expensive commercial software or high-performance computing clusters.

1.7 Limitations

The following limitations of the current study are acknowledged and should be considered when interpreting the results:

- **No experimental validation.** All compound classifications and virtual screening predictions are purely computational. No *in vitro* biochemical assay, cell-based proliferation assay, or *in vivo* pharmacological experiment was conducted within the scope of this thesis. The predicted lead compounds remain computational hypotheses requiring experimental confirmation before any biological conclusions can be drawn.
- **Bioassay data heterogeneity.** Bioactivity data sourced from ChEMBL aggregates IC₅₀ measurements originating from hundreds of independent laboratories employing diverse assay formats, cell lines, and experimental conditions. This inherent variability introduces label noise into the training dataset. While pIC₅₀ averaging and strict data filtering mitigate this to some extent, residual heterogeneity may impose a ceiling on achievable test-set performance under scaffold splitting.
- **Wild-type EGFR specificity.** No dedicated, large-scale bioactivity dataset exists for triple-mutant EGFR (Ex19del/T790M/C797S) in ChEMBL. Consequently, the model was trained on the aggregate EGFR bioactivity data of ChEMBL203, which is dominated by wild-type and single-mutant measurements. While the prospective screening of structurally novel Nepali plant scaffolds is designed to surface chemotypes distinct from existing covalent inhibitors, the model cannot explicitly predict activity against the C797S mutant form.
- **Incomplete database coverage of Nepali medicinal flora.** The virtual screening library is limited to compounds recorded for the seven target species in the COCONUT 2.0 database. Coverage is uneven across species and many traditional Nepali plant compounds remain uncharacterised and absent from public cheminformatics databases, so the screened library is not exhaustive of the phytochemical diversity of these plants.
- **Model-sensitivity of the virtual screen.** The classifiers were trained on synthetic kinase inhibitors but applied to structurally dissimilar natural products, and the resulting hit list depends strongly on the model. As reported in Chapter 4, the number of predicted actives and the identity of the top leads differ markedly between the tuned AttentiveFP, the best graph network (GCN), and the best classifier (Random Forest). No single model can be treated as authoritative; the study mitigates this by deploying the two best models and prioritising cross-model consensus leads, but only experimental assay can resolve the residual uncertainty.
- **Out-of-distribution prediction reliability.** On chemical space far from the training distribution, the weakest model (AttentiveFP) assigned near-saturated probabilities to large polyphenolic tannins that fail drug-likeness filters, and even the best models still ranked some large, non-drug-like polyphenols and triterpenoids highly—indicating that such predictions must be interpreted with caution and coupled with physicochemical filtering.
- **Two-dimensional molecular representation.** The molecular graph encodes the constitutional (2D) topology of the molecule: atom identity and bonding connectivity. Three-dimensional conformational effects, induced-fit protein-ligand binding geometry, explicit solvent interactions, and protein-mediated allosteric effects are not represented in the input features.

- **Binary classification scope.** The study frames EGFR inhibition as a binary classification problem. This formulation discards the continuous nature of the pIC_{50} scale and cannot distinguish between a moderately potent compound ($\text{pIC}_{50} = 6.1$) and a highly potent compound ($\text{pIC}_{50} = 9.5$). Continuous regression modelling of pIC_{50} values is beyond the current scope but is recommended for future work.
- **ADMET-proxy limitations.** The ADMET profiling in this study relies solely on RDKit-computed physicochemical descriptors (MW, LogP, HBD, HBA, TPSA, QED) as proxies for oral bioavailability. These filters do not capture metabolic stability, cytochrome P450 inhibition, plasma protein binding, blood-brain barrier permeability, hERG channel toxicity, or other critical ADMET parameters that determine clinical development success.

1.8 Organisation of the Thesis

The remainder of this thesis is organised as follows. **Chapter 2** provides a comprehensive review of the relevant literature, covering the biology of EGFR and NSCLC, the clinical development and resistance mechanisms of EGFR TKIs, public bioactivity databases, molecular representation methods, GNN architectures for drug discovery, traditional Nepali medicinal plants as a source of EGFR inhibitor candidates, and virtual screening approaches. **Chapter 3** describes the complete materials and methods of the five-phase computational pipeline, including data acquisition, molecular graph construction, model architectures, training protocol, hyperparameter optimisation, and the Nepali medicinal plant virtual screening workflow. **Chapter 4** presents all experimental results with tables and analytical interpretation, covering dataset characterisation, model benchmarking, hyperparameter tuning outcomes, and Nepali plant virtual screening lead prioritisation. **Chapter 5** draws the principal conclusions of the thesis and recommends directions for future experimental and computational work.

Chapter 2

Literature Review

This chapter provides a comprehensive review of the scientific literature underpinning this thesis. Section 2.1 reviews the molecular biology of EGFR and its role in NSCLC oncogenesis. Section 2.2 discusses the three generations of EGFR TKIs and their clinical limitations. Section 2.3 describes the ChEMBL bioactivity database and associated data quality challenges. Section 2.4 reviews molecular representation methods for machine learning. Section 2.5 surveys the GNN architectures employed in this study. Section 2.6 reviews the Nepali medicinal plants and natural product databases used for virtual screening. Section 2.7 reviews virtual screening methodologies. Section 2.8 synthesises the identified research gaps that motivate this thesis.

2.1 EGFR Kinase Biology and NSCLC

2.1.1 Molecular Architecture of EGFR

The Epidermal Growth Factor Receptor (EGFR), also designated ErbB1 or HER1, is a 134 kDa single-pass transmembrane receptor tyrosine kinase (RTK) encoded by the *EGFR* gene on chromosome 7p11.2. EGFR is the founding member of the ErbB receptor family (ErbB1–ErbB4), which collectively orchestrates fundamental cellular processes including proliferation, differentiation, survival, and motility (Hynes & Lane, 2005). Structurally, EGFR comprises a glycosylated **extracellular domain** (I–IV) that binds EGF-family ligands and mediates dimerisation, a single-pass **transmembrane** helix, a regulatory **juxtamembrane** segment, and a bilobed intracellular **kinase domain**. The ATP-binding cleft—formed at the N-/C-lobe interface and lined by the hinge, P-loop, activation loop, and DFG motif—is the target of all small-molecule EGFR inhibitors, and a carboxy-terminal tail bears the autophosphorylation sites (Tyr1068, Tyr1173, and others) that recruit downstream effectors.

Under normal conditions EGFR is autoinhibited; ligand binding drives homo- or heterodimerisation with other ErbB members (HER2–HER4), relieving autoinhibition and triggering *trans*-autophosphorylation that activates the RAS–RAF–MEK–ERK (MAPK), PI3K–AKT–mTOR, STAT3/5, and PLC- γ signalling cascades—collectively driving proliferation, survival, and differentiation (Siegel et al., 2023).

2.1.2 Oncogenic EGFR Mutations in NSCLC

In non-small cell lung cancer (NSCLC), somatic gain-of-function mutations within the EGFR kinase domain render the receptor constitutively active in a ligand-independent manner, producing continuous, dysregulated stimulation of the pro-survival and proliferative pathways described above (Rotow & Bivona, 2023). This oncogenic signalling drives uncontrolled tumour growth, resistance to apoptosis, angiogenesis, and metastatic dissemination.

The two most clinically prevalent sensitising EGFR mutations, which together account for 85–90% of all EGFR-mutant NSCLC, are:

- **Exon 19 deletions (Ex19del):** A heterogeneous group of in-frame deletions of 9–24 base pairs within exon 19, most commonly removing the LREA motif (residues 747–750) from the α C-helix of the kinase N-lobe. These deletions stabilise the active conformation of the kinase domain by repositioning the α C-helix, reducing the activation energy for catalysis. Ex19del accounts for approximately 45–50% of EGFR-mutant NSCLC.
- **L858R point mutation:** A leucine-to-arginine substitution at position 858 within the activation loop (A-loop) of the kinase domain. This mutation disrupts the autoinhibitory network involving Leu858, Asp855, and Phe856 of the DFG motif, shifting the equilibrium from the inactive to the active kinase conformation. L858R accounts for approximately 40% of EGFR-mutant NSCLC.

Rare sensitising mutations (<5% combined) include exon 18 G719X, exon 20 S768I, and exon 21 L861Q. These display differential sensitivity to approved TKIs and often require alternative treatment strategies.

The prevalence of EGFR mutations in NSCLC exhibits a profound ethnic disparity. In Caucasian populations, EGFR mutations are found in approximately 10–15% of NSCLC cases, whereas in East Asian populations (Chinese, Japanese, Korean, and related groups), the prevalence reaches 40–50% (Leonetti et al., 2019). This epidemiological pattern reflects underlying differences in the frequency of germline polymorphisms at the *EGFR* locus that predispose to somatic mutation acquisition. Importantly, this disparity has direct clinical relevance for South and East Asia, including Nepal, where EGFR-targeted therapy represents the most impactful precision oncology intervention available, yet access to diagnostic testing and novel therapeutic agents remains significantly constrained.

2.2 Three Generations of EGFR TKIs and Resistance Mechanisms

2.2.1 First-Generation Reversible TKIs

The clinical development of EGFR tyrosine kinase inhibitors began with the regulatory approval of gefitinib (Iressa, AstraZeneca) in 2003 and erlotinib (Tarceva, Genentech/Roche) in 2004. Both agents are reversible, ATP-competitive small molecule inhibitors that occupy the EGFR ATP-binding cleft through a combination of a hinge-region hydrogen bond (to the backbone amide nitrogen of Met793), hydrophobic contacts with the ribose pocket, and van der Waals interactions with the hydrophobic regions I and II of the binding site. Crystal structures of EGFR bound to gefitinib (PDB: 2ITY) and erlotinib (PDB: 1M17) have confirmed these binding modes at the atomic level (Stamos et al., 2002).

Landmark clinical trials established the superiority of first-generation TKIs over platinum-doublet chemotherapy in patients with sensitising EGFR mutations. The IPASS trial (gefitinib vs. carboplatin/paclitaxel) and EURTAC trial (erlotinib vs. platinum doublet) demonstrated objective response rates of 55–75% and median

progression-free survival (PFS) of 9–13 months—unprecedented outcomes for a molecularly unselected metastatic NSCLC population.

However, acquired resistance develops in virtually all patients within 9–13 months. The most common resistance mechanism (50–60% of cases) is the **T790M gatekeeper mutation** (Thr790Met) in exon 20 of the EGFR kinase domain (Leonetti et al., 2019). This methionine substitution acts through two complementary mechanisms: (i) steric clash with the quinazoline scaffold of first-generation inhibitors due to the bulkier methionine side chain, and (ii) a 100-fold increase in the binding affinity of the kinase for ATP, effectively out-competing the reversible TKI. Additional resistance mechanisms include MET amplification (~5–15%), HER2 amplification (~5–10%), PIK3CA mutation (~5%), and histological transformation to small cell carcinoma (~5%) (Leonetti et al., 2019).

2.2.2 Second-Generation Irreversible TKIs

The T790M-mediated resistance to first-generation TKIs motivated the development of second-generation irreversible inhibitors: afatinib (Gilotrif, Boehringer Ingelheim) and dacomitinib (Vizimpro, Pfizer). These agents form a covalent Michael addition bond with the thiol group of Cys797 at the rim of the ATP-binding cleft, providing kinetically stable, irreversible target engagement. As pan-ErbB inhibitors, they also inhibit HER2 and HER4 kinase activity.

Afatinib demonstrated superiority over platinum-doublet chemotherapy in the LUX-Lung 3 and LUX-Lung 6 trials, and superior PFS versus gefitinib in LUX-Lung 7, establishing it as a first-line option for EGFR-mutant NSCLC. Dacomitinib similarly improved PFS over gefitinib in the ARCHER 1050 trial. However, the irreversible pan-ErbB inhibition profile of second-generation agents produces significant dose-limiting on-target toxicity—primarily grade 3/4 diarrhoea, skin rash, stomatitis, and paronychia—arising from inhibition of wild-type EGFR in the gastrointestinal epithelium and skin (Leonetti et al., 2019). This toxicity profile restricts clinical dosing to sub-therapeutic levels that are insufficient to overcome T790M-driven resistance in the majority of patients.

2.2.3 Third-Generation Covalent TKIs and the C797S Resistance Problem

The clinical unmet need created by T790M resistance drove the development of third-generation mutant-selective covalent TKIs, led by osimertinib (Tagrisso, AstraZeneca; AZD9291). Osimertinib was rationally designed through structure-based drug design to selectively engage T790M-mutant EGFR while sparing wild-type EGFR. Its indole-based acrylamide scaffold binds the EGFR ATP cleft and forms an irreversible covalent bond with Cys797 through a Michael addition, while its selectivity for mutant over wild-type EGFR is achieved through differential binding geometries and kinetics rather than steric exclusion (Rotow & Bivona, 2023).

The AURA3 trial established osimertinib as the standard second-line therapy for T790M-positive NSCLC after first-generation TKI failure. Subsequently, the FLAURA trial demonstrated osimertinib's superiority over first-generation TKIs as first-line therapy (median PFS 18.9 vs. 10.2 months; overall survival 38.6 vs. 31.8 months), establishing it as the unambiguous standard of care for EGFR-mutant NSCLC regardless

of prior treatment. Additional advantages include excellent CNS penetration (~75% brain metastasis response rate) and an improved safety profile compared to afatinib.

Despite these remarkable clinical achievements, acquired resistance to osimertinib emerges in virtually all patients within 9–18 months (Leonetti et al., 2019). The dominant resistance mechanism is the **C797S mutation** (Cys797Ser), which substitutes the cysteine residue required for covalent TKI binding with a serine, abolishing the Michael addition reaction. C797S is detected in 20–40% of post-osimertinib biopsies and is of particular clinical importance when it occurs in *cis* with T790M on the same allele, generating the **triple-mutant EGFR** (Ex19del/T790M/C797S and L858R/T790M/C797S) that is resistant to *all three generations* of approved EGFR TKIs:

- First-generation TKIs cannot overcome T790M.
- Second-generation TKIs cannot overcome T790M steric clash at therapeutic doses.
- Third-generation TKIs cannot bind due to loss of the Cys797 covalent anchor.

Additional post-osimertinib resistance mechanisms include acquired MET amplification (~15%), HER2 amplification (~5%), RAS/RAF/MAP2K mutations (~10%), EGFR amplification, EGFR C797 allosteric mutations (e.g., G796S, L792 mutations), and histological transformation to SCLC or squamous cell carcinoma (Rotow & Bivona, 2023). As of 2026, no approved therapeutic agent effectively addresses triple-mutant EGFR. Investigational fourth-generation strategies—including reversible, mutant-selective non-covalent inhibitors (e.g., BLU-945) and allosteric inhibitors that bind outside the ATP pocket (e.g., EAI045)—are in preclinical and early clinical development, but none is yet approved and the durability of response remains uncertain. This makes the discovery of structurally novel, non-covalent EGFR inhibitor chemotypes—including those derived from natural products—the highest clinical priority in NSCLC drug development.

2.3 Bioactivity Databases and the ChEMBL Database

2.3.1 Overview of the ChEMBL Database

The ChEMBL database, maintained by EMBL-EBI, is the largest curated public repository of bioactive small molecules and their quantitative activities (Mendez et al., 2019), comprising over 2.3 million compounds and 20 million bioactivity measurements across more than 14,000 targets, curated primarily from the medicinal-chemistry literature. The EGFR entry (target ID: ChEMBL203) is among its most extensively annotated targets, reflecting decades of drug-discovery activity around this oncology target.

2.3.2 Bioactivity Endpoints and pIC₅₀ Conversion

The primary bioactivity endpoint used in this study is the **half-maximal inhibitory concentration** (IC₅₀), defined as the concentration of a compound required to inhibit the target activity by 50% under specified assay conditions. IC₅₀ values are expressed in molar concentration units (typically nanomolar, nM) and span many orders of magnitude across structurally diverse compounds.

For machine learning purposes, IC_{50} values are routinely converted to the logarithmic pIC_{50} scale by taking the negative base-ten logarithm of the molar concentration, so that higher pIC_{50} values correspond to more potent compounds. This transformation compresses the large dynamic range of IC_{50} values (spanning 6–8 orders of magnitude) into a more tractable numerical range (approximately 4–11 for drug-like compounds), improves the normality of the activity distribution, and is the standard practice in QSAR modelling. A threshold of $pIC_{50} \geq 6.0$ (corresponding to an IC_{50} of $\leq 1 \mu M$) is the conventional medicinal chemistry criterion for classifying a compound as a “hit” or “active” in kinase inhibitor discovery (Wu et al., 2018).

2.3.3 Data Quality Challenges

Utilising ChEMBL data for machine learning presents several important quality challenges that must be addressed during data curation (Wu et al., 2018):

- **Assay heterogeneity:** IC_{50} values from hundreds of laboratories use different assay formats and conditions, which can shift absolute values by 1–2 orders of magnitude and add label noise.
- **Censored measurements:** Many values are censored ($>$ or $<$); standard practice retains only exact (“=”) measurements.
- **Replicate measurements:** Repeated tests of the same compound are aggregated by the median pIC_{50} to limit outlier influence.
- **Structural redundancy:** Salt forms and notation differences are resolved by RD-Kit canonicalisation and largest-fragment salt stripping.
- **Activity cliffs:** Structurally similar compounds with very different activities challenge both fingerprint- and graph-based models.

2.4 Molecular Representation for Machine Learning

2.4.1 Hand-Crafted Fingerprint Descriptors

Molecular fingerprints are fixed-length binary or integer vectors that encode the presence or count of predefined structural features within a molecule. They represent the dominant molecular representation paradigm in traditional cheminformatics and remain strong baselines for bioactivity prediction tasks.

2.4.1.1 Extended Connectivity Fingerprints (ECFP)

The Extended Connectivity Fingerprint (ECFP) (Rogers & Hahn, 2010), also known as the Morgan fingerprint (after its algorithmic basis in the Morgan algorithm for canonical atom numbering), is the most widely used molecular fingerprint in cheminformatics. ECFP encodes circular atomic neighbourhoods through an iterative procedure:

1. Each atom is assigned an initial integer identifier based on six atomic invariants (atomic number, degree, number of hydrogens, charge, isotope, and ring membership).

2. At each iteration r (radius), each atom’s identifier is updated by hashing together its current identifier with the identifiers of its immediate neighbours (corresponding to atoms within r bonds).
3. All identifiers generated at each iteration up to radius r are collected and folded into a bit vector of specified length (typically 1024 or 2048 bits) using a hash function.

ECFP4 (radius = 2, 2048 bits) is the de facto standard, encoding circular substructures up to 4 bonds in diameter. Its advantages include computational efficiency ($O(N)$ in molecular size), interpretability of individual bits as specific substructural features, and strong empirical performance across diverse QSAR benchmarks. Limitations include: (i) fixed-length encoding discards variable-size topological patterns not captured within the predefined radius; (ii) bit collisions in folded fingerprints reduce information content; (iii) bond-level information (bond type, conjugation, stereochemistry) is only coarsely encoded; and (iv) the representation is not differentiable, precluding end-to-end learning of task-optimal molecular representations.

2.4.1.2 Other Descriptor Types

Beyond fingerprints, molecular property prediction has traditionally relied on physico-chemical descriptors (e.g., RDKit 2D and 3D descriptors, MACCS keys, DRAGON descriptors), shape-based 3D pharmacophore fingerprints, and continuous molecular embeddings derived from SMILES string language models.

For the purposes of this study, ECFP4 fingerprints are used for both traditional machine learning baselines. **Random Forest (RF)** is an ensemble of 500 decision trees with balanced class weighting and `max_features='sqrt'`, providing a well-calibrated, sample-efficient baseline for medium-sized bioactivity datasets. The **Multi-Layer Perceptron (MLP)** baseline is a three-layer neural network (2048→512→128→1) operating on the same ECFP4 inputs, trained with BCEWithLogitsLoss and early stopping. The MLP bridges the gap between fingerprint-based and graph-based deep learning, allowing direct attribution of performance differences to the graph representation rather than to deep learning capacity in general.

2.4.2 Graph-Based Molecular Representations

The fundamental insight motivating the graph-based approach is that a molecule is inherently a labelled graph: atoms are nodes with associated chemical properties, and bonds are edges with associated chemical characteristics. This isomorphism allows molecules to be fed directly into graph processing algorithms without information-losing pre-encoding steps (Gilmer et al., 2017).

A molecular graph is defined by three components:

- A set of **nodes**, where each node represents one heavy atom and carries a node feature vector describing that atom’s chemical properties.
- A set of **edges**, where each edge represents one chemical bond and carries an edge feature vector describing that bond’s properties.

- The **bonding connectivity**, recording which atoms are bonded to which.

The dimensionality and content of the node and edge feature vectors directly determine the chemical information available to the model. In this work, a comprehensive 29-dimensional node feature vector and 12-dimensional edge feature vector were designed to capture the chemical information most relevant to kinase inhibitor classification, as detailed in Chapter 3.

2.4.3 Graph Neural Networks: The Message-Passing Framework

The general message-passing neural network (MPNN) framework (Gilmer et al., 2017) provides a unified description of most GNN architectures. In this framework, each atom maintains a hidden representation that is refined layer by layer through two operations, illustrated in Figure 2:

Aggregation: each atom collects “messages” from its bonded neighbours, where each message is a learned function of the central atom’s representation, the neighbour’s representation, and the connecting bond features. The messages are then combined using a permutation-invariant operator (sum, mean, or max) so that the result does not depend on the arbitrary ordering of neighbours.

Update: each atom updates its own hidden representation by combining its previous state with the aggregated message, typically through a small multilayer perceptron or a gated recurrent unit.

After several message-passing layers, a **readout function** aggregates all the atom-level representations into a single graph-level embedding, which is then passed to a task-specific head (for example, a fully connected classifier for binary activity prediction).

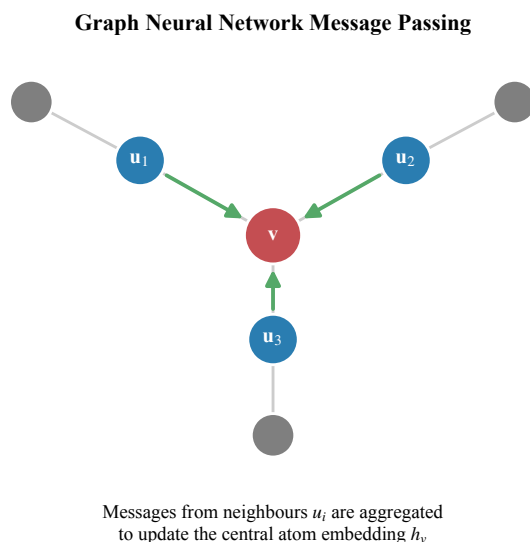


Figure 2: One step of GNN message passing: atom v aggregates learned messages from its bonded neighbours u_1, u_2, u_3 .

The key advantages of the MPNN framework for molecular property prediction are: (i) the representation is learned end-to-end from bioactivity data, requiring no expert-designed feature engineering; (ii) arbitrary atom- and bond-level information can be

incorporated via the node and edge feature vectors; (iii) the framework naturally handles molecules of varying size without fixed-length encoding; and (iv) the learned representations capture both local chemical environments (from lower layers) and global molecular topology (from higher layers through the readout).

2.5 Graph Neural Network Architectures for Drug Discovery

2.5.1 Graph Convolutional Network (GCN)

The Graph Convolutional Network (GCN) (Kipf & Welling, 2017) introduced the concept of spectral graph convolution through a computationally tractable first-order approximation. At each layer, every atom’s representation is updated by taking a degree-normalised average of the (linearly transformed) representations of its bonded neighbours together with its own, followed by a non-linear activation. The normalisation uses the atom degrees (with self-loops added) so that atoms with many bonds do not dominate the aggregation.

The GCN architecture has several notable limitations in the context of molecular property prediction. First, the degree-normalised aggregation weights all neighbouring atoms equally, without learning which chemical neighbours contribute most to a given property. Second, standard GCN implementations use node features only and do not incorporate edge features, discarding bond-level chemical information. Third, GCN is limited in its expressive power: it cannot distinguish certain pairs of non-isomorphic graph structures that differ only in their higher-order topology, limiting its ability to discriminate complex pharmacophoric patterns (Xu et al., 2019). Despite these limitations, GCN provides a strong and computationally efficient baseline and demonstrated competitive AUROC performance in the TDC ADMET benchmarks (Huang et al., 2021).

2.5.2 Graph Attention Network (GAT)

The Graph Attention Network (GAT) (Veličković et al., 2018) replaces the GCN’s fixed degree-based averaging with learnable attention coefficients: each neighbour’s contribution to an atom’s update is weighted by a learned score computed from the two atoms’ representations. Several parallel attention “heads” are used—concatenated in intermediate layers and averaged in the final one—and the GATv2 extension (Brody et al., 2022) fixes the “static attention” limitation of the original formulation by conditioning the weights dynamically on both the query and key nodes. This lets the network focus on the atoms most relevant to activity, such as the hinge-binding nitrogens of the quinazoline pharmacophore over solvent-exposed substituents.

2.5.3 Graph Isomorphism Network (GIN)

The Graph Isomorphism Network (GIN) (Xu et al., 2019) was developed from a theoretical analysis of the representational limits of GNNs within the message-passing framework. Xu et al. proved that the expressive power of any message-passing GNN is bounded by the Weisfeiler-Leman (WL) graph isomorphism test, and demonstrated that GIN achieves this theoretical upper bound. The GIN layer updates each atom by *sum-*

ming its neighbours’ representations, adding the atom’s own representation (scaled by a small learnable factor), and passing the result through a multilayer perceptron.

The critical theoretical insight is that sum aggregation preserves multiset information about neighbour features (unlike mean or max aggregation, which can conflate distinct neighbourhood configurations), making GIN maximally expressive among message-passing GNNs.

The GINEConv (Hu et al., 2020) extension incorporates edge features by adding a transformed bond feature vector to each neighbour’s representation before the summation, so that bond-level chemistry informs the aggregation.

Despite its theoretical superiority, GIN does not always outperform architecturally simpler GNNs on molecular property prediction benchmarks, suggesting that the WL-level expressivity gains are primarily relevant for highly symmetric graph structures that are uncommon in drug-like molecules. Empirically, GIN has shown strong performance on graph classification tasks in the TU graph benchmarks and MoleculeNet (Wu et al., 2018).

2.5.4 AttentiveFP

AttentiveFP (Xiong et al., 2020) is a GNN purpose-built for molecular property prediction, combining two mechanisms. At the **atom level**, attention weights modulating each neighbour’s contribution are computed from the two atoms’ representations *together with the connecting bond’s feature vector*, natively integrating edge chemistry into the attention. At the **molecule level**, an iterative readout refines a single molecular embedding: a context vector, initialised from the mean atom representation, repeatedly attends over all atoms and is updated by a Gated Recurrent Unit (GRU) over several timesteps, capturing progressively more global structure. AttentiveFP reaches near state-of-the-art performance on several MoleculeNet benchmarks (Jiang et al., 2021; Xiong et al., 2020), which is why it was chosen as the architecture for hyperparameter optimisation here—although the benchmark in Chapter 4 ultimately found it the weakest of the five graph networks on the scaffold test set.

2.5.5 GraphSAGE

GraphSAGE (Graph SAmple and aggreGatE) (Hamilton et al., 2017) is an inductive framework that learns aggregation functions from sampled local neighbourhoods rather than the full adjacency matrix. Each atom forms the mean of its neighbours’ representations, concatenates this with its own previous representation, and passes the result through a learnable transformation—preserving the atom’s identity alongside its local context. Mean aggregation is robust to high-degree atoms, and the inductive formulation generalises naturally to novel scaffolds, which is directly relevant under Bemis-Murcko evaluation. Here GraphSAGE uses BatchNorm and concatenated mean+max pooling, following the same readout as GCN.

2.6 Nepali Medicinal Plants and Natural Product Databases

2.6.1 Traditional Medicinal Plants in Nepal

Nepal harbours remarkable biodiversity, hosting approximately 6,500 flowering plant species within a geographic transect spanning tropical lowlands (~60 m a.s.l.) to alpine zones (>5,000 m a.s.l.). Of these, over 700 species have documented medicinal uses in traditional Ayurvedic and Tibetan medical systems. Several Nepali medicinal plants have been ethnopharmacologically associated with anticancer activity and are candidates for systematic computational evaluation as potential EGFR inhibitor sources.

Seven species were selected on the basis of documented bioactivity, coverage in COCONUT 2.0, and relevance to oncological conditions:

- *Swertia chirayita* (Chirayito) — xanthones and secoiridoids;
- *Rhododendron arboreum* (Lali Gurans) — triterpenoids and flavonoids;
- *Nardostachys jatamansi* (Jatamansi) — sesquiterpenes;
- *Terminalia chebula* (Haritaki) — hydrolysable tannins, a Triphala constituent;
- *Berberis aristata* (Chutro) — the alkaloid berberine, with reported EGFR-pathway modulation;
- *Ocimum sanctum* (Tulsi) — eugenol and ursolic/rosmarinic acids, with reported EGFR-inhibitory activity;
- *Tinospora cordifolia* (Giloy) — diterpenoids and isoquinoline alkaloids.

All seven have ethnopharmacologically documented anti-inflammatory or antiproliferative use.

2.6.2 The COCONUT 2.0 Database

The **CO**llection of **O**pen **N**atural **P**rodUc**T**s (COCONUT) is the largest freely available database of natural product structures, curated and maintained by the Steinbeck group at Friedrich Schiller University Jena (Sorokina et al., 2021). COCONUT 2.0 contains over 400,000 unique natural product structures drawn from more than 50 public natural product databases, each associated with organism taxonomy, literature references, and precomputed molecular descriptors. Structures are provided as canonical SMILES and InChI strings with CAS registry numbers where available.

For this study, COCONUT 2.0 was used to retrieve SMILES for compounds annotated to the seven Nepali plant species of interest using organism taxonomy matching. Retrieved compounds were then subjected to the same preprocessing pipeline as the ChEMBL training data (salt removal, SMILES canonicalisation, $MW \leq 1000$ Da, RDKit validation) to ensure featurisation consistency.

2.7 Virtual Screening in Drug Discovery

2.7.1 Overview of Virtual Screening

Virtual screening (VS) is a computational strategy employed at the early stage of drug discovery to prioritise a large library of candidate compounds for experimental testing, based on predicted interaction with a biological target or predicted biological activity. VS substantially reduces the number of compounds that need to be synthesised and assayed experimentally, focusing experimental resources on the most promising candidates (Walters & Murcko, 2020).

Two principal paradigms of VS are distinguished:

- **Structure-based virtual screening (SBVS):** Requires a three-dimensional crystal or cryo-EM structure of the target protein. Molecular docking algorithms (AutoDock Vina, Glide, GOLD) predict the binding pose and estimate the binding affinity of each candidate compound within the target’s binding site. SBVS is accurate but computationally intensive and requires high-resolution structural data.
- **Ligand-based virtual screening (LBVS):** Does not require target structural knowledge. A model trained on known active and inactive compounds (e.g., from ChEMBL) is used to score and rank external compound libraries by predicted activity probability. LBVS is computationally scalable to millions of compounds and is particularly powerful when high-quality bioactivity training data is available.

This thesis implements a **ligand-based virtual screening** pipeline using its two best-performing trained classifiers—the best graph neural network (GCN) and a Random Forest baseline—as the scoring functions, and prioritising leads by cross-model consensus rather than relying on any single model.

2.7.2 GNN-Based Virtual Screening

The landmark application of GNN-based virtual screening in drug discovery was the discovery of **halicin** by Stokes et al. (Stokes et al., 2020) at the Collins Laboratory (MIT). A directed-message passing neural network (DMPNN) was trained on 2,335 compounds from the Drug Repurposing Hub to predict broad-spectrum antibiotic activity. The trained model was applied to score over 100 million compounds from the ZINC15 database; halicin (SU-3327, a c-Jun N-terminal kinase inhibitor) was identified as a top-ranked hit and subsequently validated as a potent, broad-spectrum antibiotic in mouse models of *Acinetobacter baumannii* infection. This study demonstrated that GNN-based LBVS can identify structurally novel bioactive compounds that escape detection by conventional similarity searches.

In the kinase inhibitor domain, GNN-based VS has been applied to several targets including EGFR, CDK2, and VEGFR2, typically achieving hit rates of 10–50% in subsequent experimental validation—substantially higher than the 0.1–1% hit rates typical of unguided high-throughput screening (Sun et al., 2020).

2.7.3 Natural Product Libraries and Nepali Medicinal Plants

A key methodological choice in virtual screening is the source and composition of the screening library. This thesis employs a **natural product library** strategy: bioactive compounds from seven traditional Nepali medicinal plant species are retrieved from the COCONUT 2.0 database and used as the screening library. The biological and ethnopharmacological rationale is well-established: plants such as *Swertia chirayita*, *Berberis aristata*, and *Ocimum sanctum* have centuries of documented use in the treatment of cancer-related symptoms and have yielded structurally diverse secondary metabolites—alkaloids, flavonoids, terpenes, and tannins—that represent structurally distinct chemotypes compared to synthetic kinase inhibitors, potentially offering novel scaffolds capable of engaging the EGFR ATP-binding pocket through unconventional binding modes. This strategy directly connects computational drug discovery to the biodiversity resources and traditional knowledge of Nepal, supporting affordable and regionally accessible cancer therapeutic research.

2.7.4 ADMET Properties and Lead Prioritisation

Virtual screening hit lists require secondary filtering by predicted ADMET (Absorption, Distribution, Metabolism, Excretion, Toxicity) properties to prioritise compounds with the greatest potential for clinical development (Walters & Murcko, 2020). In this study, ADMET-proxy properties were computed using RDKit as physicochemical surrogates:

- **Lipinski’s Rule of Five (Ro5):** $MW \leq 500$ Da, $\text{LogP} \leq 5$, $\text{HBD} \leq 5$, $\text{HBA} \leq 10$ as necessary (not sufficient) conditions for oral bioavailability (Lipinski, 2001).
- **Quantitative Estimate of Drug-likeness (QED):** A composite drug-likeness score (range 0–1) integrating MW, LogP, HBD, HBA, TPSA, number of aromatic rings, number of rotatable bonds, and the presence of structural alerts, calibrated to reflect the desirability profile of approved oral drugs (Bickerton et al., 2012).
- **Topological Polar Surface Area (TPSA):** A molecular surface area descriptor correlated with oral absorption and blood-brain barrier permeability. Values $\leq 140 \text{ \AA}^2$ are associated with adequate gastrointestinal absorption.

2.8 Research Gap

A systematic analysis of the reviewed literature reveals three persistent and interrelated methodological gaps in the application of Graph Neural Networks to EGFR inhibitor discovery. These gaps collectively limit the clinical utility and scientific rigour of existing computational approaches:

1. **Evaluation Bias from Random Data Splitting.** The majority of published GNN models for EGFR or related kinase bioactivity prediction employ random train/test splitting (Jiang et al., 2021; Malik et al., 2025). This practice allows molecules with near-identical Bemis-Murcko scaffolds to appear simultaneously in training and test partitions, creating an optimistic bias in reported performance metrics. A model that achieves 0.90 AUROC under random splitting may perform

significantly worse when deployed on a genuinely novel scaffold—precisely the scenario encountered in prospective virtual screening. Scaffold-based splitting, which groups all molecules sharing the same core framework into a single partition, provides a more realistic and demanding assessment of model generalisation (Wu et al., 2018).

2. **Incomplete Molecular Featurisation.** Published GNN models for EGFR inhibitor classification frequently employ minimal node feature sets (e.g., atom type only, or a small number of atomic properties) and rarely incorporate explicit, multi-dimensional bond-level (edge) feature vectors (Malik et al., 2025). Bond features encoding stereochemistry, conjugation, and aromaticity carry significant pharmacological information—for example, the aromatic bonds of the quinazoline hinge-binder scaffold and the acrylamide double bond of covalent inhibitors—that is discarded or only coarsely approximated by node-only models or ECFP fingerprints.
3. **Lack of Prospective Screening on Natural Product and Regionally Relevant Compound Libraries.** Most academic GNN papers in the EGFR and kinase inhibitor domain report performance metrics on held-out test sets but do not deploy the trained model in a prospective virtual screening pipeline applied to natural product or ethnopharmacological compound collections (Malik et al., 2025; Sun et al., 2020). For South Asian research communities where EGFR-mutant NSCLC is prevalent but novel targeted therapies are expensive and inaccessible, virtual screening of traditional medicinal plant compounds represents a high-impact translational application that is entirely absent from the computational EGFR inhibitor literature.

Table 1 formally positions this thesis against representative published GNN studies for EGFR inhibitor prediction.

Table 1: This thesis versus representative GNN-based EGFR studies across three methodological dimensions (✓ addressed; ◦ partial; × absent).

Study	Scaffold Split	Rich Features	Natural Product Screening
DeepEGFR (Malik et al., 2025)	×	◦	×
GCN QSAR (Jiang et al., 2021)	×	◦	×
AttentiveFP (Xiong et al., 2020)	×	✓	×
Hu et al. (Hu et al., 2020)	◦	✓	×
This Thesis	✓	✓	✓

This thesis addresses all three gaps through its two objectives, within a single reproducible open-source pipeline. To close the evaluation and featurisation gaps (1–2), it **benchmarks graph neural networks**—built with comprehensive 29-dimensional node and 12-dimensional edge features and tuned via Optuna—**against traditional machine-learning baselines** under a strict Bemis-Murcko scaffold split. To close the screening

gap (3), it **deploys the best-performing models** in a prospective virtual screen of Nepali medicinal plant compounds from COCONUT 2.0 with ADMET-proxy prioritisation.

Chapter 3

Materials and Methods

3.1 System Flow Diagram of the Computational Framework

3.1.1 Overview of the Computational Framework

The overall methodology follows a systematic five-phase pipeline as illustrated in Figure 3. All code was implemented in Python 3.14 using exclusively open-source libraries and executed on a macOS system (CPU).

3.2 Data Acquisition

3.2.1 Target Selection

The primary target for bioactivity data retrieval was the Epidermal Growth Factor Receptor (EGFR), indexed in ChEMBL as target identifier CHEMBL203. EGFR was selected as the model target because it is a clinically validated oncology target in NSCLC, and CHEMBL203 aggregates the largest publicly available collection of quantitative EGFR inhibition bioactivity measurements.

3.2.2 Bioactivity Data Retrieval

Bioactivity data was retrieved programmatically from the ChEMBL database using the `chembl-webresource-client` Python library, which queries the then-current ChEMBL release through the REST API (retrieved May 2026). The query was restricted to compounds satisfying the following criteria:

- **Standard type:** IC₅₀ (half-maximal inhibitory concentration)
- **Standard relation:** Exact value only (=)

Records with missing SMILES strings, missing IC₅₀ values, or non-exact relations (>, <, ≥, ≤) were excluded. IC₅₀ values reported in ChEMBL for this target are in nanomolar (nM) units, as assumed throughout the pIC₅₀ conversion. The raw dataset contained approximately 19,000 compound-activity pairs prior to deduplication and cleaning.

3.2.3 Data Cleaning and Labelling

The raw IC₅₀ values (in nanomolar units) were converted to the logarithmic pIC₅₀ scale by taking the negative base-ten logarithm of the molar concentration, so that more potent compounds receive higher pIC₅₀ scores. Binary activity labels were then assigned using the standard medicinal chemistry threshold: compounds with a pIC₅₀ of at least 6.0 were labelled *active*, and those below 6.0 were labelled *inactive*. This threshold corresponds

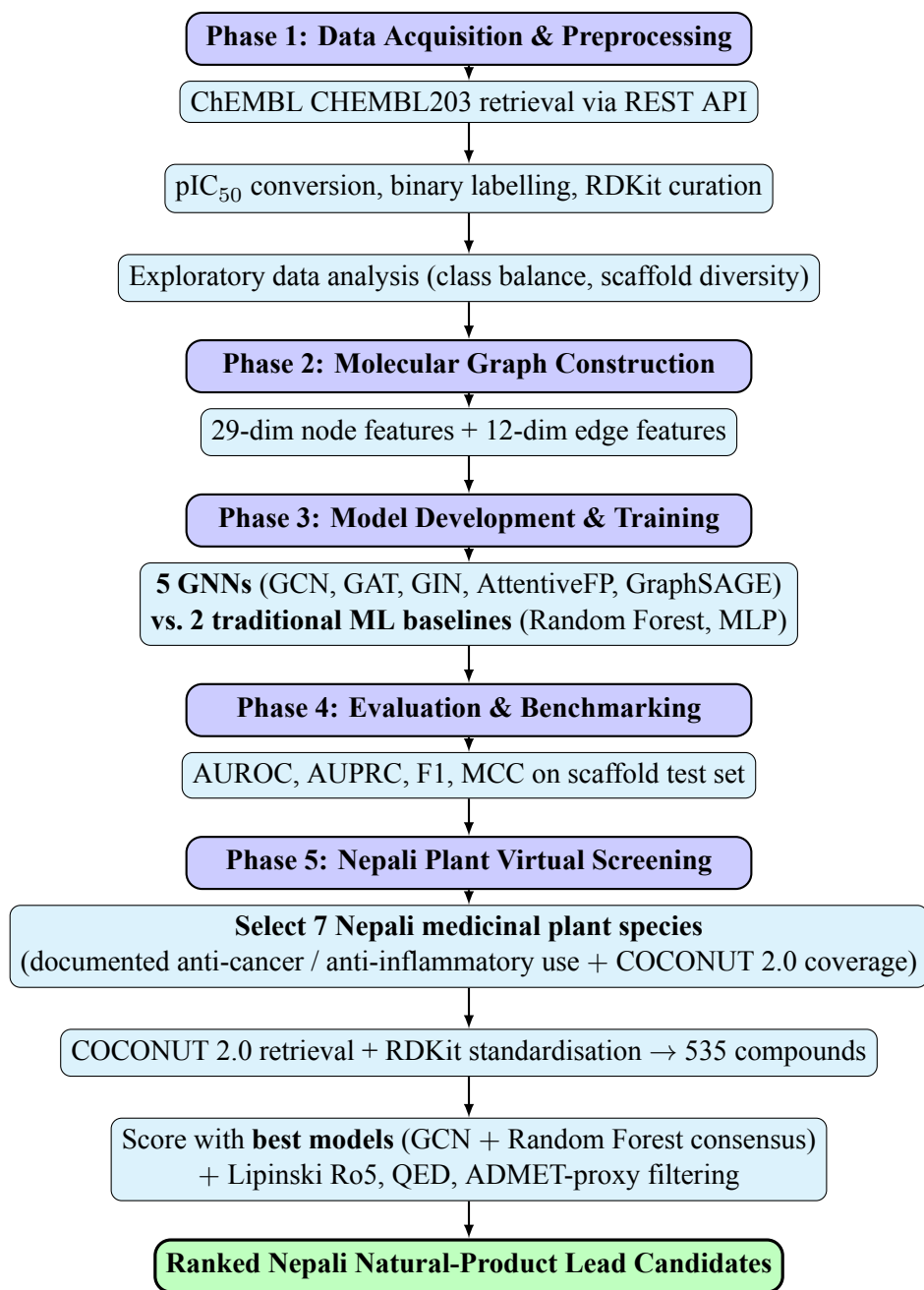


Figure 3: Five-phase pipeline: benchmarking five GNNs against traditional ML baselines, then screening Nepali medicinal-plant compounds with the best models.

to an IC_{50} of $\leq 1 \mu M$. For compounds with multiple reported measurements, the median pIC_{50} was used to obtain a single robust value per molecule.

3.2.4 Molecular Standardisation and Filtering

Molecular standardisation was performed using RDKit (version 2026.3.2):

- **Salt removal:** The largest fragment was retained for each compound.
- **Canonicalisation:** SMILES strings were canonicalised to ensure unique molecular representation.
- **Molecular weight filter:** Compounds with $MW > 1000$ Da were excluded.
- **Deduplication:** Duplicate canonical SMILES were collapsed using the median pIC_{50} .

The final curated dataset comprised **10,523 unique compounds**, distributed as 7,792 actives (74.0%) and 2,731 inactives (26.0%).

3.2.5 Exploratory Data Analysis

Prior to molecular graph construction, an exploratory data analysis (EDA) step was performed on the cleaned dataset to characterise its statistical structure and inform modelling decisions. Using RDKit and pandas, the analysis quantified the class balance, the distribution of pIC_{50} potency values, the ranges of the physicochemical properties (molecular weight, LogP), and the Bemis-Murcko scaffold diversity (number of unique scaffolds and the proportion of singleton scaffolds). This characterisation, reported in Chapter 4, both confirms the integrity of the curated dataset and motivates the use of scaffold-based splitting, given the long-tailed scaffold distribution observed.

3.3 Molecular Graph Construction

3.3.1 Graph Representation

Each molecule was converted to an undirected graph using PyTorch Geometric (version 2.7.0), where every node represents an atom and every edge represents a chemical bond (the molecule-to-graph conversion is illustrated in Figure 1). Each atom node carries a 29-dimensional feature vector and each bond edge a 12-dimensional feature vector, detailed below.

3.3.2 Node Features

Each atom was represented by a 29-dimensional feature vector encoding:

- **Atom type** (one-hot, 10 dim): C, N, O, S, F, Cl, Br, I, P, Unknown
- **Atom degree** (one-hot, 6 dim): 0, 1, 2, 3, 4, ≥ 5
- **Formal charge** (1 dim): integer value

- **Hydrogen count** (one-hot, 5 dim): 0, 1, 2, 3, ≥ 4
- **Hybridisation** (one-hot, 5 dim): SP, SP2, SP3, SP3D, SP3D2
- **Aromaticity** (binary, 1 dim)
- **Ring membership** (binary, 1 dim)

3.3.3 Edge Features

Each bond was represented by a 12-dimensional edge feature vector encoding:

- **Bond type** (one-hot, 4 dim): Single, Double, Triple, Aromatic
- **Conjugation** (binary, 1 dim)
- **Ring membership** (binary, 1 dim)
- **Stereochemistry** (one-hot, 6 dim): STEREOANY, STEREOZ, STEREOE, STEREOCIS, STEREOTRANS

3.4 Data Splitting

To ensure rigorous evaluation on genuinely novel chemical scaffolds, a deterministic **Bemis-Murcko Scaffold Split** was implemented using RDKit. The Bemis-Murcko algorithm extracts the core macrocyclic framework of each molecule by retaining ring systems, linker atoms, and chain atoms connected to rings while removing side chains. Compounds sharing identical scaffold hashes were allocated exclusively to one split partition. The dataset was partitioned as follows:

Table 2: Dataset partitioning using the Bemis-Murcko scaffold split.

Partition	Proportion	Compounds	Unique Scaffolds
Training	80%	8,418	majority
Validation	10%	1,052	held-out
Test	10%	1,053	held-out

3.5 Machine Learning Model Development

3.5.1 Traditional Baseline Models

Two traditional machine learning baselines were implemented using scikit-learn (version 1.8.0) trained on 2048-bit ECFP4 (Morgan) fingerprints (radius = 2) computed by RDKit:

- **Random Forest (RF)**: An ensemble of 500 decision trees with `class_weight='balanced'` to account for the 74:26 class imbalance. Hyperparameters included `max_features='sqrt'` and `min_samples_leaf=1`.

- **Multi-Layer Perceptron (MLP):** A three-layer neural network (2048→512→128→1) with BatchNorm and Dropout(0.3) at each hidden layer, trained on ECFP4 fingerprints with BCEWithLogitsLoss (pos_weight), Adam optimiser, and early stopping (patience = 10) on an internal 90/10 validation split. This evaluates whether deep learning on fingerprint features alone offers benefits beyond the Random Forest ensemble.

3.5.2 Graph Neural Network Architectures

Five GNN architectures were implemented using PyTorch Geometric. Each learns atom embeddings by repeatedly aggregating information from bonded neighbours, pools them into a single molecular vector, and outputs an activity probability through a final sigmoid layer,

$$\hat{p} = \sigma(z) = \frac{1}{1 + e^{-z}} \in (0, 1). \quad (1)$$

The architectures differ only in how each atom aggregates information from its neighbours, described below.

3.5.2.1 Graph Convolutional Network (GCN)

The GCN propagates information using degree-normalised spectral convolution. At each layer, every atom embedding is updated by averaging the transformed embeddings of its bonded neighbours and itself, where the averaging weights are normalised by the degrees of the participating atoms (with added self-loops). This symmetric normalisation prevents high-degree atoms from dominating the representation. Three convolutional layers were stacked, each followed by batch normalisation, a ReLU non-linearity, and dropout.

3.5.2.2 Graph Attention Network (GAT)

The GAT replaces the fixed normalisation of the GCN with learnable attention weights: for each atom, the contribution of every neighbour is weighted by an attention coefficient computed from the two atoms’ embeddings and the connecting bond features, then normalised across all neighbours. This lets the network learn which bonds and substructures are most relevant to activity. Four attention heads were used in the intermediate layers (their outputs concatenated) and a single averaged head in the final layer. Bond (edge) features were supplied directly to the attention mechanism through the GATv2 formulation.

3.5.2.3 Graph Isomorphism Network (GIN)

GIN is the most theoretically expressive message-passing scheme within the Weisfeiler–Lehman framework. Rather than averaging, each atom embedding is updated by summing its neighbours’ embeddings (preserving multiset information that mean or max aggregation would discard), adding the atom’s own embedding, and passing the result through a two-layer multilayer perceptron. The GINEConv variant used here extends this update to incorporate the 12-dimensional bond feature vectors.

3.5.2.4 GraphSAGE

GraphSAGE (Hamilton et al., 2017) updates each atom by combining its own previous embedding with the mean of its neighbours’ embeddings and passing the combination through a learnable transformation. Because the update depends only on local neighbourhood aggregation rather than the global graph structure, GraphSAGE is inherently inductive and generalises naturally to unseen molecules. Three SAGEConv layers with batch normalisation and ReLU activation are used, followed by concatenated global mean and max pooling and a linear classification head. GraphSAGE was included to provide an inductive neighbourhood-sampling baseline that is evaluated under the same scaffold split conditions as the attention-based models.

3.5.2.5 AttentiveFP

AttentiveFP implements a two-level attention mechanism. At the atom level, a graph attention layer aggregates neighbourhood information weighted by learned atom–atom attention scores. At the molecule level, a Gated Recurrent Unit (GRU)-based readout iteratively refines a single global molecular embedding over several timesteps, allowing the model to focus on the substructures most predictive of activity. Crucially, AttentiveFP natively accepts both the atom and bond feature vectors without preprocessing, making it uniquely suited to exploit the full 29-dimensional node and 12-dimensional edge feature specification.

3.5.3 Training Protocol

All GNN models were trained with the following protocol:

- **Loss function:** Binary cross-entropy with logits incorporating a positive class weight (the ratio of negative to positive training examples, approximately 0.34) to address the class imbalance.
- **Optimiser:** Adam with its default momentum parameters.
- **Scheduler:** ReduceLROnPlateau with reduction factor 0.5 and patience 10 epochs.
- **Early stopping:** Training halted if validation AUROC did not improve for 20 consecutive epochs (30 for the tuned AttentiveFP).
- **Batch size:** 64 for default training; selected by Optuna for the tuned AttentiveFP.
- **Readout:** Global mean and max pooling were concatenated to form the graph-level representation.

3.5.4 Model Evaluation Metrics

Models were evaluated on the scaffold-held-out test set with five complementary metrics. Four derive from the confusion-matrix counts of true positives (TP), false positives

(FP), true negatives (TN), and false negatives (FN) at the $\hat{p} = 0.5$ decision threshold:

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN}, & \text{Precision} &= \frac{TP}{TP + FP}, & \text{Recall} &= \frac{TP}{TP + FN}, \\ & & & & (2) \\ \text{F1} &= \frac{2TP}{2TP + FP + FN}, & \text{MCC} &= \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}. \end{aligned}$$

F1 is the harmonic mean of precision and recall, while the Matthews correlation coefficient (MCC) $\in [-1, 1]$ uses all four counts and is robust to class imbalance; it is reported as the primary balanced metric. The two threshold-independent metrics are the area under the receiver-operating-characteristic curve (AUROC) of the true-positive rate $\text{TPR} = TP/(TP + FN)$ against the false-positive rate $\text{FPR} = FP/(FP + TN)$ —equivalently, the probability that a random active is ranked above a random inactive, $\text{AUROC} = \Pr(\hat{p}_+ > \hat{p}_-)$ —and the area under the precision–recall curve (AUPRC), which is more informative than AUROC under the 74:26 class imbalance.

3.5.5 Hyperparameter Tuning

Bayesian hyperparameter optimisation was performed for the AttentiveFP model using the Optuna framework (version 4.8.0) with the Tree-structured Parzen Estimator (TPE) sampler. The search space encompassed:

Table 3: Hyperparameter search space for AttentiveFP Optuna optimisation.

Hyperparameter	Type	Range / Options
hidden_channels	Categorical	{64, 128, 256}
num_layers	Integer	2–4
num_timesteps	Integer	2–4
dropout	Float (log)	0.1–0.5
batch_size	Categorical	{32, 64, 128}
learning_rate	Float (log)	10^{-4} – 10^{-2}
weight_decay	Float (log)	10^{-5} – 10^{-3}

The objective function to maximise was validation AUROC. Pruning of unpromising trials was performed using the Optuna MedianPruner.

3.6 Nepali Medicinal Plant Virtual Screening

3.6.1 Plant Species and Compound Library Curation

A virtual screening library was assembled from the COCONUT 2.0 natural product database (Sorokina et al., 2021) by filtering for compounds annotated to seven traditional Nepali medicinal plant species:

Table 4: Nepali medicinal plant species used for virtual screening.

Species	Common Name	Traditional Use
<i>Swertia chirayita</i>	Chirayito	Fever, liver conditions
<i>Rhododendron arboreum</i>	Lali Gurans	Anti-inflammatory
<i>Nardostachys jatamansi</i>	Jatamansi	Neurological, cardiac
<i>Terminalia chebula</i>	Haritaki	Antioxidant, Triphala
<i>Berberis aristata</i>	Chutro	Anticancer (berberine)
<i>Ocimum sanctum</i>	Tulsi	Anticancer, anti-EGFR
<i>Tinospora cordifolia</i>	Giloy/Gurjo	Immunomodulatory

Retrieved compounds were subjected to the same preprocessing pipeline as the ChEMBL training data: salt removal (largest fragment retained), SMILES canonicalisation, molecular weight filter ($MW \leq 1000$ Da), removal of invalid SMILES, and deduplication by canonical SMILES. ADMET-proxy descriptors (MW, LogP, HBD, HBA, TPSA, QED) were computed using RDKit for all retained compounds.

3.6.2 Inference and Prioritisation

To avoid basing lead selection on any single model—and in particular to avoid the weakest classifier—the library was scored with the study’s two best-performing models: the best graph neural network (GCN) and the best overall classifier (Random Forest). For the GCN, each compound was featurised using the identical 29-dimensional node and 12-dimensional edge graph specification as the training data and passed through a forward inference to produce $\hat{p}_i = P(\text{active} \mid G_i)$. For the Random Forest, each compound was encoded as a 2048-bit ECFP4 fingerprint and scored with the trained ensemble. Predictions from both models were retained so that consensus (agreement between the two best models) could be assessed. The tuned AttentiveFP screen was also computed for comparison, exposing its over-confidence relative to the two best-calibrated models.

Compounds were ranked by predicted activity probability $\hat{p}_i$ under each model. The top leads were further assessed by:

- **Lipinski Rule of Five compliance:** $MW \leq 500$ Da, $\text{LogP} \leq 5$, $\text{HBD} \leq 5$, $\text{HBA} \leq 10$ (necessary condition for oral bioavailability).
- **QED score:** A composite drug-likeness score targeting $\text{QED} \geq 0.40$.
- **TPSA:** Values $\leq 140 \text{ \AA}^2$ for adequate oral absorption.

The final outputs are prioritised ranked lists for each best model (`results/nepali_leads_rf.csv` and `results/nepali_leads_gcn.csv`) of Nepali natural-product-derived compounds with computed ADMET-proxy properties, representing actionable candidates for experimental EGFR inhibition assays.

Chapter 4

Results and Discussion

This chapter presents the complete experimental results of the five-phase computational pipeline. Section 4.1 characterises the curated EGFR bioactivity dataset. Section 4.2 evaluates and compares all seven classification models. Section 4.3 reports Nepali medicinal plant virtual screening outcomes and lead prioritisation. Section 4.4 provides an integrated discussion of all findings.

4.1 Data Distribution Analysis

4.1.1 Dataset Composition and Class Balance

The curated dataset comprised **10,523 unique EGFR-bioactive compounds** retrieved from ChEMBL (ChEMBL203) following strict quality filtering. The binary label distribution exhibited a moderate class imbalance: **7,792 active compounds** (74.0%, $\text{pIC}_{50} \geq 6.0$) and **2,731 inactive compounds** (26.0%, $\text{pIC}_{50} < 6.0$). This imbalance is inherent to bioactivity databases and reflects the practical reality that published compounds are disproportionately those that demonstrated measurable activity in assay screening campaigns.

Table 5 summarises the dataset composition across the three scaffold-split partitions.

Table 5: Dataset composition and class distribution across scaffold-split partitions.

Partition	Total	Active	Inactive	Active (%)
Training	8,418	6,292	2,126	74.7
Validation	1,052	743	309	70.6
Test	1,053	757	296	71.9
Total	10,523	7,792	2,731	74.0

The class balance remained broadly consistent across all three partitions (74.7% / 70.6% / 71.9%), with the held-out validation and test partitions being marginally less active-enriched. This is expected under a strict Bemis-Murcko scaffold split, in which the large shared scaffold families are concentrated in the training set and the held-out partitions are dominated by novel singleton scaffolds. The overall class balance is shown in Figure 4.

4.1.2 pIC_{50} Distribution

The pIC_{50} distribution across the full dataset exhibited a bimodal profile. The active subpopulation was centred around $\text{pIC}_{50} \approx 7.5\text{--}8.0$ (corresponding to IC_{50} values of

EGFR Dataset Class Balance

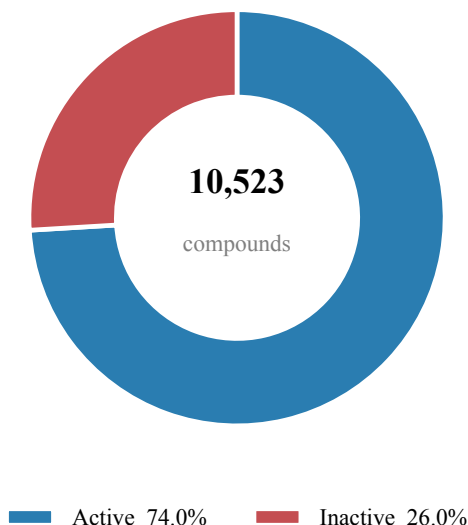


Figure 4: Class balance of the curated EGFR dataset: 74.0% active and 26.0% inactive compounds across 10,523 unique molecules.

10–32 nM), reflecting the high-potency compounds typically selected for further optimisation in medicinal chemistry campaigns. The inactive subpopulation showed a peak near $\text{pIC}_{50} \approx 5.2\text{--}5.5$, representing weak binders at the micro-molar level. The binary threshold at $\text{pIC}_{50} = 6.0$ ($\text{IC}_{50} = 1\text{ }\mu\text{M}$) was placed in a relatively low-density valley between these two modes, providing a chemically meaningful separation rather than an arbitrary cut-point. This is consistent with the standard medicinal chemistry convention that defines a “hit” as a compound with measurable activity at or below $1\text{ }\mu\text{M}$. The bimodal structure and the placement of the decision threshold are shown in Figure 5.

4.1.3 Physicochemical Property Analysis

Analysis of the RDKit-computed physicochemical properties of the curated dataset revealed that the majority of compounds occupied drug-like chemical space. Key observations are summarised in Table 6.

Table 6: Summary statistics of physicochemical properties of the curated dataset.

Property	Min	Median	Mean	Max
Molecular Weight (Da)	65.4	478.0	475.0	991.7
LogP	−2.0	4.5	4.5	13.1
H-Bond Donors	0	2	2.2	10
H-Bond Acceptors	0	7	6.6	17
TPSA ($\text{\AA}^2$)	0.0	96.3	95.2	273.0

The median molecular weight of 478 Da and median LogP of 4.5 sit close to the upper limits of Lipinski’s Rule of Five (Ro5), consistent with the physicochemical characteris-

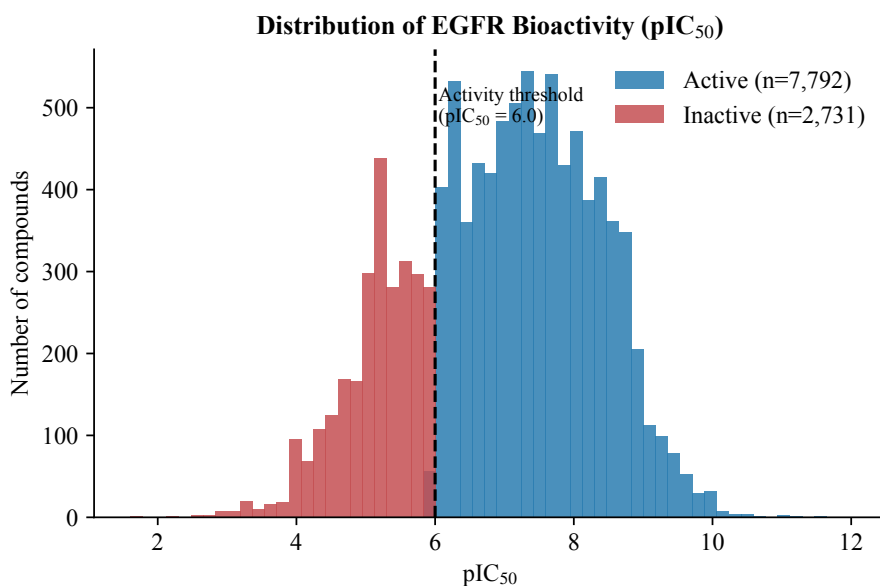


Figure 5: Distribution of pIC_{50} for the curated dataset; the two activity modes are split by the $\text{pIC}_{50} = 6.0$ ($1\ \mu\text{M}$) threshold (dashed).

tics of ATP-competitive kinase inhibitors, which frequently occupy the upper boundary of drug-like chemical space. Approximately **47%** of the compounds were fully Lipinski-compliant ($\text{MW} \leq 500\ \text{Da}$, $\text{LogP} \leq 5$, $\text{HBD} \leq 5$, $\text{HBA} \leq 10$); the remainder most often exceed the LogP or molecular weight thresholds, reflecting the lipophilic, higher-mass character typical of this inhibitor class. The molecular weight and lipophilicity distributions are shown in Figure 6.

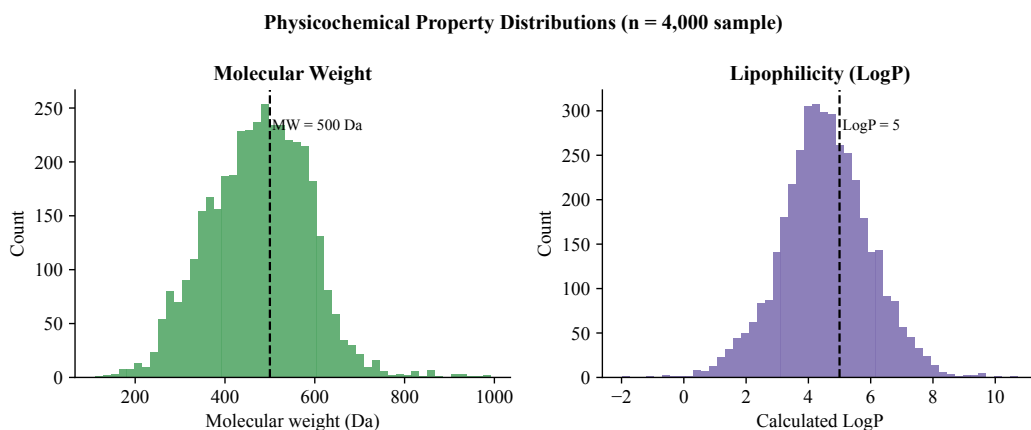


Figure 6: Molecular weight (left) and LogP (right) distributions for the curated dataset; dashed lines mark the Lipinski thresholds (500 Da, 5).

4.1.4 Scaffold Diversity Analysis

The Bemis-Murcko scaffold analysis identified **3,693 unique scaffolds** across the full dataset. Of these, 2,418 (65.5%) were singletons, comprising only a single compound. The most frequently occurring scaffold, observed in 572 compounds, corresponds to the 4-anilinoquinazoline core, which is the defining pharmacophore of first- and second-

generation EGFR TKIs including gefitinib, erlotinib, and lapatinib. Several further scaffolds each comprising over one hundred compounds include the pyrrolopyrimidine core characteristic of certain second-generation and third-generation inhibitors. This scaffold-level analysis confirms that the dataset represents a structurally diverse yet pharmacophore-focused chemical space. Figure 7 shows the most frequent scaffolds and the overall split between singleton and multi-compound scaffolds, which motivates the use of scaffold-based data splitting.

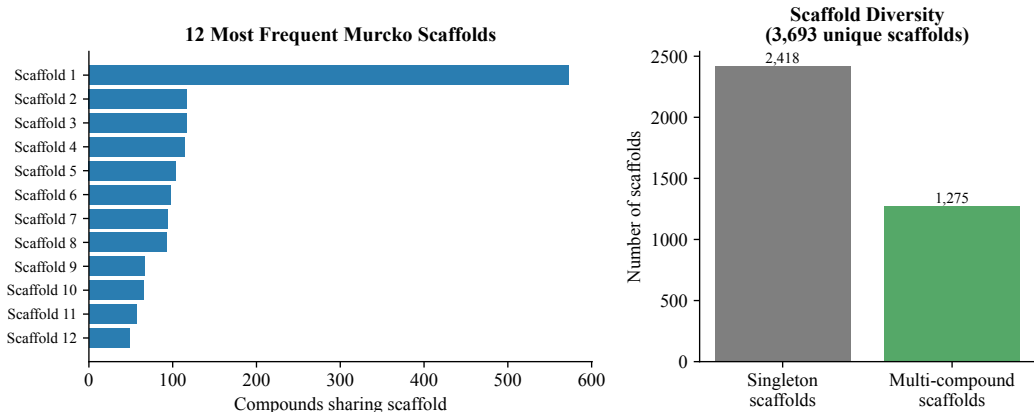


Figure 7: Bemis-Murcko scaffold diversity: most frequent scaffolds (left) and the singleton vs. multi-compound split (right).

4.2 Machine Learning Model Performance

4.2.1 Individual Model Performance on the Scaffold Test Set

All seven models were trained under identical scaffold split conditions and evaluated on the held-out scaffold test set of 1,053 compounds. Results are presented in Table 7. The classification threshold was set at $\hat{p} = 0.5$ for all threshold-dependent metrics (F1-Score, MCC, Accuracy).

Table 7: Performance comparison of all models on the scaffold-held-out test set (threshold = 0.50 for F1, MCC, Accuracy).

Model	Type	AUROC	AUPRC	F1	MCC	Accuracy
Random Forest	ML + ECFP4	0.8827	0.9496	0.8762	0.5072	0.8129
MLP	ML + ECFP4	0.8547	0.9354	0.8629	0.5281	0.8053
GCN	GNN	0.8454	0.9332	0.7877	0.4707	0.7312
GraphSAGE	GNN	0.8229	0.9179	0.8070	0.4706	0.7474
GIN	GNN	0.8217	0.9163	0.8487	0.4865	0.7863
GAT	GNN	0.8162	0.9176	0.7689	0.4473	0.7123
AttentiveFP	GNN	0.8087	0.9108	0.8254	0.3989	0.7521

Figure 8 visualises these results across the four headline metrics, making the relative ordering of the seven models directly comparable.

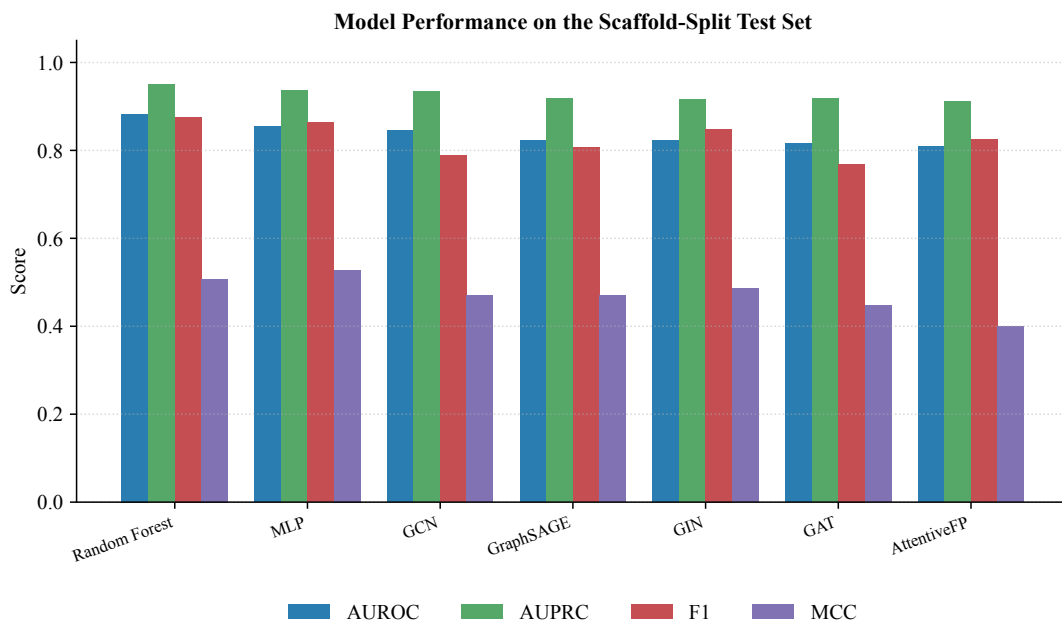


Figure 8: Performance of all seven models on the scaffold test set; fingerprint baselines lead, with GCN strongest among the GNNs.

4.2.1.1 Random Forest Performance

Random Forest achieved the highest test-set AUROC (0.8827) and AUPRC (0.9496) among all models, together with the strongest F1-Score (0.8762); the MLP baseline attained the highest MCC (0.5281), narrowly ahead of Random Forest (0.5072). The strong performance of the two fingerprint-based baselines is consistent with established observations in cheminformatics benchmarks: ensemble and shallow neural methods built on ECFP4 fingerprints are highly competitive on bioactivity datasets of this scale ($\sim 10,000$ compounds), particularly when the training and test chemical spaces share fingerprint-level structural similarity. The high AUPRC value (0.9496) is especially significant given the 74:26 class imbalance, indicating that Random Forest maintains high precision at high recall thresholds.

4.2.1.2 Graph Neural Network Performance

Among the five GNN architectures at default settings, **GCN achieved the highest AUROC (0.8454)**, followed by GraphSAGE (0.8229), GIN (0.8217), GAT (0.8162), and AttentiveFP (0.8087). Notably, the simpler convolution- and sampling-based architectures outperformed the more elaborate attention-based models on this metric—a common observation, since high-capacity, molecule-specific architectures such as AttentiveFP are highly sensitive to hyperparameter choices and tend to underperform simpler models until properly tuned (Section 4.2.2). The per-architecture characteristics are:

- **GCN** provided the strongest and most stable default GNN, its degree-normalised convolution generalising well to novel scaffolds without extensive tuning.

- **GraphSAGE** performed comparably (AUROC 0.8229), with its concatenated self-and-neighbour update giving robust inductive generalisation on molecular graphs of this scale.
- **GIN** achieved the strongest F1-Score (0.8487) and MCC (0.4865) among the GNNs; its sum aggregation preserves neighbourhood multiset information that benefits threshold-dependent classification.
- **GAT** did not translate its learned attention weights into a higher AUROC than GCN at default settings, indicating that the additional parameters require tuning to pay off.
- **AttentiveFP**, despite its sophisticated two-level attention and native 12-dimensional edge-feature integration, recorded the lowest default AUROC (0.8087) and MCC (0.3989). This points to substantial untapped capacity that motivates the dedicated hyperparameter optimisation reported in Section 4.2.2.

4.2.1.3 GNN vs. Traditional ML Baselines

The performance gap between the fingerprint baselines and all default GNN models (AUROC difference: 0.037–0.074 relative to Random Forest) reflects a well-documented phenomenon in molecular machine learning: GNNs require larger training datasets and careful tuning to overcome the strong inductive bias provided by ECFP4 fingerprints. Notably, the MLP baseline (AUROC 0.8547), which applies deep learning to the same ECFP4 fingerprints, outperformed all five default GNN models, confirming that the fixed fingerprint representation—rather than deep learning capacity alone—drives the advantage of the baselines. The ECFP4 representation was refined through decades of cheminformatics research specifically for molecular activity prediction, while GNN architectures must learn to identify relevant substructures from first principles. This gap is not closed by hyperparameter optimisation: tuning AttentiveFP (Section 4.2.2) improved its validation AUROC but left its test AUROC essentially unchanged, so on the strict scaffold test set the fingerprint baselines remained ahead of every graph neural network.

4.2.2 Hyperparameter Tuning Results

Bayesian hyperparameter optimisation was performed over 20 Optuna trials using the Tree-structured Parzen Estimator sampler, with the objective of maximising validation AUROC. The optimal hyperparameter configuration identified for AttentiveFP is presented in Table 8.

Table 8: Optimal AttentiveFP hyperparameters identified by Optuna Bayesian optimisation over 20 trials, with the resulting performance.

Hyperparameter	Default Value	Optimal Value
hidden_channels	128	128
num_layers	2	3
num_timesteps	2	2
dropout	0.30	0.129
batch_size	64	64
learning_rate	0.001	2.61×10^{-3}
weight_decay	0.0	1.10×10^{-5}
Best validation AUROC (20 trials)		0.8632
Tuned test AUROC (cf. default 0.8087)		0.7970

The optimisation raised the best *validation* AUROC to **0.8632**, a modest improvement that lifts AttentiveFP into the middle of the field on the validation metric. On the held-out scaffold *test* set, however, the tuned model achieved an AUROC of only **0.7970**—essentially unchanged from the untuned model (0.8087), still the lowest of the GNNs, and well below the Random Forest (0.8827) and MLP (0.8547) fingerprint baselines. In other words, hyperparameter tuning improves the validation-set fit but does not translate into better test-set generalisation, and the gap to the fingerprint-based baselines is not closed on this strict scaffold-split task—a realistic outcome consistent with the well-documented difficulty of out-performing tuned fingerprint models on medium-sized bioactivity datasets.

The selected hyperparameters are nonetheless informative:

- **num_layers = 3**: A third message-passing layer enables atoms to exchange information across up to three bonds, covering the spatial extent of most pharmacophoric motifs (the default used only two).
- **dropout = 0.129**: A lower dropout than the default (0.30) was preferred, indicating that the model benefits from retaining more of its representational capacity on this dataset.
- **learning_rate = 2.61×10^{-3}** : A higher learning rate than the default (10^{-3}) accelerated convergence within the per-trial epoch budget.
- **weight_decay = 1.10×10^{-5}** : A small non-zero L2 penalty (absent by default) provided mild regularisation against scaffold-specific overfitting.
- **hidden_channels, num_timesteps, and batch_size** were retained at their default values (128, 2, and 64), indicating these were already near-optimal for the task.

4.3 Nepali Medicinal Plant Virtual Screening Results

4.3.1 Screening Library Composition

The Nepali medicinal plant screening library was assembled from the COCONUT 2.0 database by filtering for compounds annotated to seven traditional Nepali plant species, then applying the same RDKit standardisation pipeline as the ChEMBL training data. Filtering and standardisation returned **535 unique natural product compounds**. Crucially, to ensure a genuinely *prospective* screen, the library was then de-duplicated against the ChEMBL bioactivity data: **eight compounds whose canonical SMILES already appear in the ChEMBL EGFR training data** (including the flavones **apigenin** and **kaempferol**, and **quercetin**, ellagic acid, pyrogallol, and pyrocatechol) were removed, since these are not novel natural-product candidates but known compounds the classifiers may already have seen during training. This leaves **527 genuinely unseen compounds** that were screened, distributed across the seven species as summarised in Table 9.

Table 9: Composition of the screened Nepali plant library after removing eight ChEMBL-overlapping compounds (COCONUT 2.0).

Plant Species	Compound Class	Compounds
<i>Nardostachys jatamansi</i>	Sesquiterpenes, coumarins	199
<i>Terminalia chebula</i>	Hydrolysable tannins	192
<i>Tinospora cordifolia</i>	Diterpenes, alkaloids	71
<i>Swertia chirayita</i>	Xanthones, secoiridoids	23
<i>Berberis aristata</i>	Isoquinoline alkaloids	19
<i>Rhododendron arboreum</i>	Triterpenoids, flavonoids	15
<i>Ocimum sanctum</i>	Phenylpropanoids, terpenoids	8
Total	—	527

The library is dominated by *Nardostachys jatamansi* (199 compounds) and *Terminalia chebula* (192), reflecting their extensive phytochemical characterisation in COCONUT, whereas *Ocimum sanctum* (8) and *Rhododendron arboreum* (15) are comparatively under-represented—an illustration of the uneven database coverage of Nepali medicinal flora. The distribution across the seven species is shown in Figure 9.

4.3.2 Physicochemical Distribution of the Screening Library

Before scoring, the physicochemical property distribution of the 527-compound COCONUT library was characterised and compared against the ChEMBL EGFR training set of Section 4.1, because the two distributions determine how far the classifiers must *extrapolate*. The comparison is summarised in Table 10 and visualised in Figure 10.

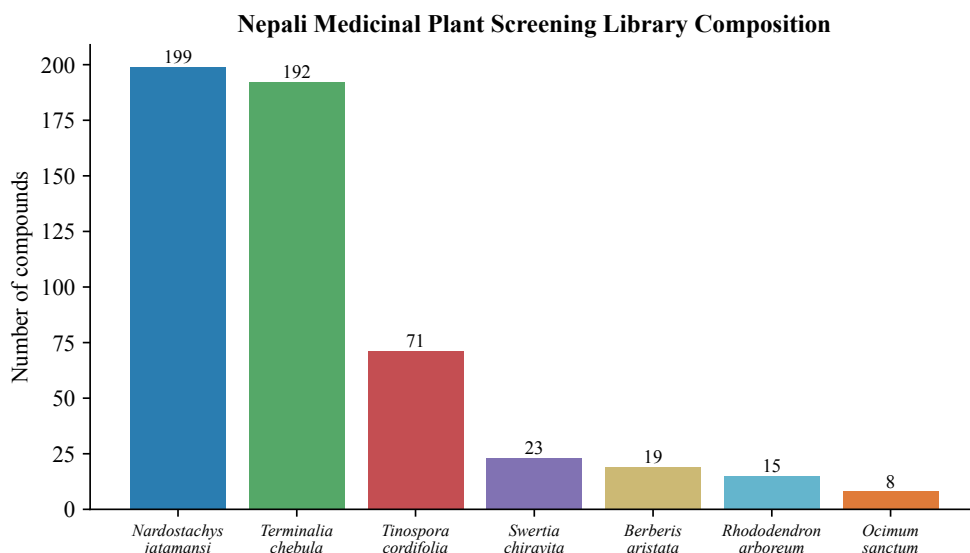


Figure 9: Screened Nepali plant library by species (527 genuinely unseen compounds; 535 retrieved minus eight in ChEMBL).

Table 10: Physicochemical profile of the 527-compound Nepali library versus the 10,523-compound ChEMBL training set.

Property (median)	COCONUT library (527)	ChEMBL training (10,523)
Molecular weight (Da)	341.4	478.0
LogP	2.5	4.5
H-bond donors	2	2
H-bond acceptors	4	7
TPSA ($\text{\AA}^2$)	66.8	96.3
QED	0.4	0.4
Lipinski-compliant (%)	53.7	46.6

The natural-product library is chemically distinct from the training data on every size- and polarity-related axis. Its median molecular weight (341.4 Da) is **137 Da lighter** than the ChEMBL median (478.0 Da), its median LogP (2.5) is a **full two log-units less lipophilic** (4.5), and its median TPSA ($66.8 \text{ \AA}^2$) is substantially lower ($96.3 \text{ \AA}^2$). A larger fraction of the natural products are fully Lipinski-compliant (53.7% vs. 46.6%), consistent with the generally lower molecular weight of plant secondary metabolites relative to optimised synthetic inhibitors. Figure 10 shows that the two molecular-weight and lipophilicity distributions overlap only partially: the COCONUT library carries a pronounced low-mass, low-LogP shoulder (small terpenoids and phenylpropanoids) that is essentially absent from the training set, while also retaining a tail of very large (MW > 900 Da) hydrolysable tannins from *Terminalia chebula*.

This distribution shift is the mechanistic explanation for the over-confidence documented later in this section: the classifiers were trained almost entirely on heavier, more lipophilic quinazoline-type inhibitors and must *extrapolate* to score the lighter,

more polar natural products, so their predicted probabilities on this library are not calibrated and must be treated as rankings rather than calibrated likelihoods.

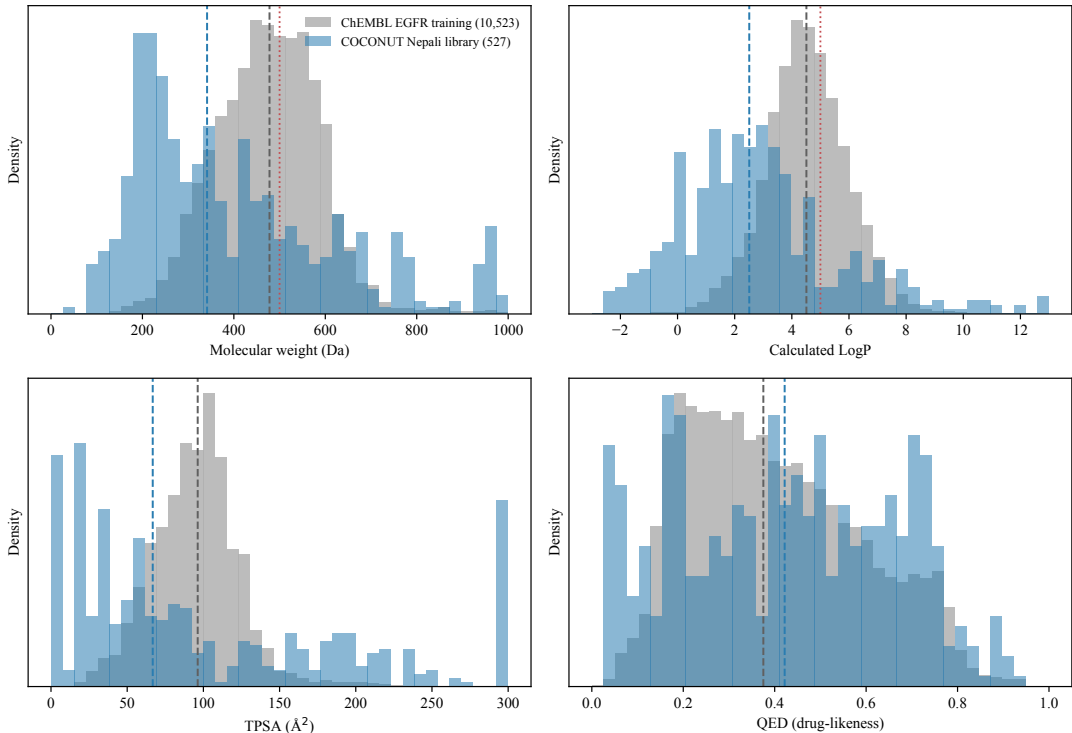


Figure 10: Physicochemical distributions (MW, LogP, TPSA, QED) of the Nepali library (blue) versus the ChEMBL training set (grey); dashed lines mark medians.

4.3.3 Multi-Model Screening Funnel and Hit Identification

A preliminary screen scored the library with the tuned AttentiveFP model. However, AttentiveFP was in fact the *weakest* classifier on the scaffold test set (Section 4.2). To avoid basing lead selection on the weakest model, the prospective screen was re-run and deployed with the study’s two *best-performing* models—the strongest graph neural network (GCN, test AUROC 0.8454) and the strongest overall classifier (Random Forest, test AUROC 0.8827)—applied to the identical 527-compound de-duplicated library. Table 11 compares the progressive screening funnel obtained under all three models.

Table 11: Nepali plant screening funnel under the three models, showing compounds retained at each activity and drug-likeness stage.

Filter Stage	AttentiveFP (0.81)	GCN (0.85)	Random Forest (0.88)
Screened compound library (de-duplicated)	527	527	527
Predicted active ($\hat{p} \geq 0.50$)	143	24	179
Predicted active ($\hat{p} \geq 0.80$)	83	1	4
Predicted active ($\hat{p} \geq 0.90$)	71	0	0
Drug-like leads ($\hat{p} \geq 0.90$, Ro5, QED ≥ 0.40)	11	0	0

The three funnels differ dramatically, and this divergence is itself a central result. The tuned AttentiveFP predicts 143 actives at $\hat{p} \geq 0.50$ and an implausible 71 compounds at $\hat{p} \geq 0.90$; the best graph network (GCN) predicts only 24 actives and *not a single* compound at $\hat{p} \geq 0.90$; and the best overall classifier (Random Forest) predicts 179 actives but, on the genuinely unseen library, likewise *no* compound at $\hat{p} \geq 0.90$. Consequently, once the eight ChEMBL-overlapping compounds are removed, *neither* best model yields a high-confidence ($\hat{p} \geq 0.90$) drug-like lead—the single Random Forest hit above $\hat{p} \geq 0.90$ in a preliminary run was apigenin, a compound already present in the training data (see Section 4.4). The abundance of apparently high-confidence hits in the AttentiveFP screen is therefore an artefact of poor probability calibration on out-of-distribution natural products by the weakest model, and is *not* reproduced by either best-calibrated model. This model-sensitivity is the principal caution of the screening campaign: virtual-screening hit lists on natural-product chemical space depend strongly on the choice of classifier, and the most trustworthy leads are those nominated by the strongest models—ideally by more than one. A second, related artefact—over-confidence on large, non-drug-like natural products—persists across models: AttentiveFP scored large hydrolysable *Terminalia chebula* tannins (e.g., pentagalloylglucose, MW $\approx$ 941 Da) at $\hat{p} \approx 1.0$, and both the GCN and the Random Forest assigned their highest raw scores to large *Swertia* and *Tinospora* triterpenoids that violate the Rule of Five, reinforcing the need to couple activity prediction with drug-likeness filtering. The three funnels are compared in Figure 11.

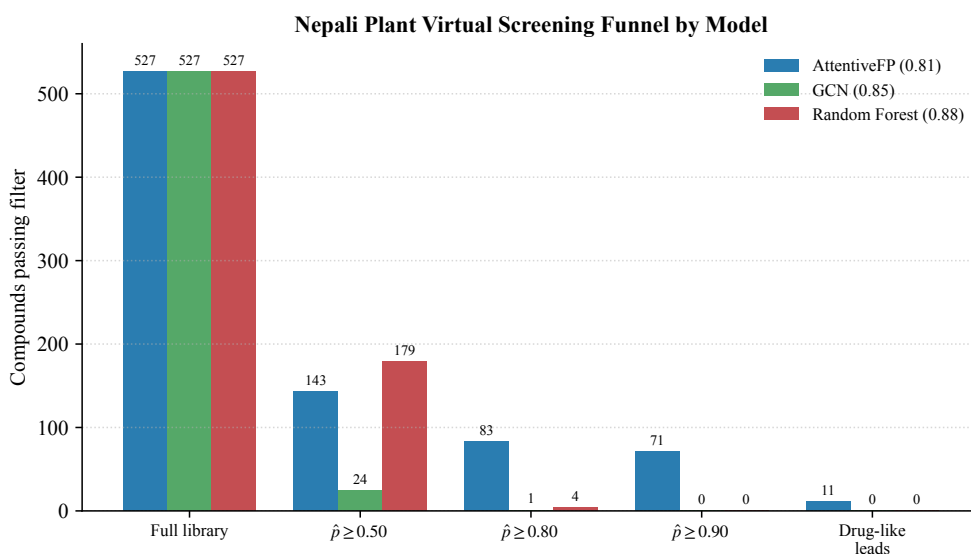


Figure 11: Screening funnel under the three models (tuned AttentiveFP, GCN, Random Forest); the sharp disagreement shows the screen’s model-sensitivity.

4.3.4 Consensus Leads from the Best-Performing Models

Because no single classifier is authoritative, lead prioritisation is based on the two best-performing models rather than the weak AttentiveFP screen. The top drug-like (Lipinski-compliant) leads nominated by the Random Forest are listed in Table 12, and those nominated by the GCN in Table 13.

Table 12: Top drug-like genuinely-unseen Nepali leads under the Random Forest, ranked by predicted activity; none reaches $\hat{p} \geq 0.80$.

Rank	Compound	$\hat{p}$	MW (Da)	LogP	QED	Ro5	Plant
1	Isogentisin	0.7282	258.2	2.37	0.655	✓	<i>S. chirayita</i>
2	Luteolin	0.6789	286.2	2.28	0.511	✓	<i>T. chebula</i>
3	Tinosporaside	0.6607	492.5	0.63	0.440	✓	<i>T. cordifolia</i>
4	Columbin	0.6180	358.4	2.53	0.613	✓	<i>T. cordifolia</i>
5	Palmarin	0.6042	374.4	1.74	0.591	✓	<i>T. cordifolia</i>
6	Nardostachysin	0.5836	430.5	2.95	0.526	✓	<i>N. jatamansi</i>

Table 13: Top drug-like Nepali leads under the GCN, ranked by predicted activity; none reaches $\hat{p} \geq 0.80$.

Rank	Compound	$\hat{p}$	MW (Da)	LogP	QED	Ro5	Plant
1	Rulepidadiene A	0.6531	202.3	4.19	0.552	✓	<i>N. jatamansi</i>
2	Nardostachysin	0.5915	430.5	2.95	0.526	✓	<i>N. jatamansi</i>
3	Palmarin	0.5532	374.4	1.74	0.591	✓	<i>T. cordifolia</i>
4	Luteolin	0.5138	286.2	2.28	0.511	✓	<i>T. chebula</i>
5	Jatamanvaltrate H	0.4995	442.5	1.70	0.405	✓	<i>N. jatamansi</i>

Two observations stand out. First, the flavone scaffold is prominent among the predicted actives, but this signal is strongly confounded by the training-set leakage identified above (Table 14): the two highest-scoring flavones under the Random Forest—apigenin ($\hat{p} = 0.913$) and kaempferol ($\hat{p} = 0.834$)—are precisely the compounds already present in the ChEMBL training data (both labelled active), so their high scores reflect *memorisation*, not prospective prediction. Once these are removed, **luteolin** is the only flavone that remains, and it is predicted active ($\hat{p} \geq 0.50$) by *both* best models (RF 0.679, GCN 0.514)—a genuinely unseen, consensus-supported candidate. Second, on the de-duplicated library both the GCN and the Random Forest assign their highest raw probabilities to large, non-drug-like *Swertia* and *Tinospora* triterpenoids, so drug-likeness filtering remains essential.

Table 14: Flavone scaffolds under the two best models with ChEMBL status; apigenin and kaempferol are leakage, luteolin the consensus lead.

Flavone	ChEMBL status	RF $\hat{p}$	GCN $\hat{p}$	Retained?
Apigenin	in training (active)	0.913	0.410	removed (leakage)
Kaempferol	in training (active)	0.834	0.320	removed (leakage)
Luteolin	not in training set	0.679	0.514	retained (consensus lead)

Figure 12 positions the Random Forest drug-like leads in the predicted-activity versus drug-likeness plane.

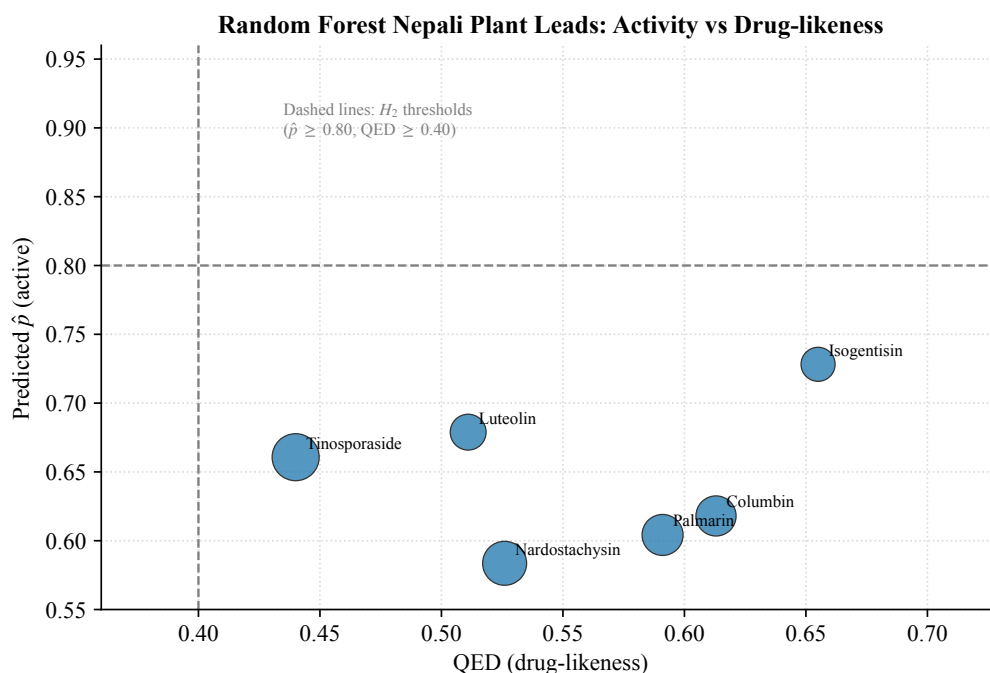


Figure 12: Random Forest drug-like leads in the predicted-activity versus QED plane; dashed lines mark the H_2 thresholds (0.80, 0.40), which none clears.

4.3.4.1 Recovered Known Actives, and the Highest-Confidence Prospective Lead

A preliminary screen of the full 535-compound library ranked two flavones highest—apigenin ($\hat{p} = 0.913$) and kaempferol ($\hat{p} = 0.834$). Both, however, are already present in the ChEMBL training data as known EGFR-active compounds, so their high scores reflect the model *recovering* known actives rather than discovering new ones. Their correct recovery is a useful positive control—the classifier does identify established EGFR-active flavones—but they are not prospective natural-product leads, and they were removed from the screening library.

On the de-duplicated (genuinely unseen) library, *no* Lipinski-compliant compound reaches $\hat{p} \geq 0.80$ under either best model. The highest-confidence unseen drug-like lead is **isogentisin** from *Swertia chirayita*, a xanthone scored by the Random Forest at $\hat{p} = 0.728$ (MW = 258.2 Da, LogP = 2.37, QED = 0.655; fully Ro5-compliant). Isogentisin is therefore a *moderate*-confidence candidate only, and—like all leads reported here—requires experimental validation.

4.3.4.2 Cross-Model Consensus Leads and Their Plant Sources

Because the two best models disagree so strongly, the most defensible lead-selection criterion is *cross-model consensus*: compounds predicted drug-like *and* active ($\hat{p} \geq 0.50$, Lipinski-compliant, QED ≥ 0.40) by *both* the Random Forest and the GCN. This intersection is deliberately conservative—it discards any lead that only one model favours, and is therefore robust to the model-sensitivity and out-of-distribution over-confidence documented above. As Figure 13 shows, of the 25 drug-like actives nominated by the Random Forest and the four by the GCN, only **three** are endorsed by both.

Drug-like predicted-active leads: cross-model consensus (Random Forest $\cap$ GCN)

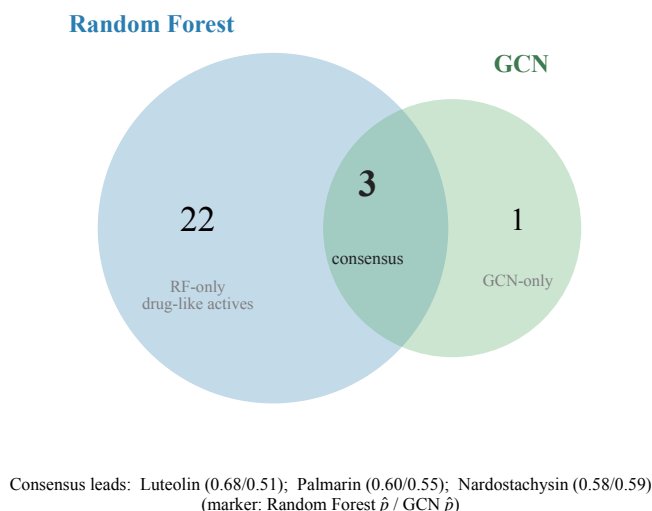


Figure 13: Cross-model consensus: drug-like actives from the Random Forest (25) and GCN (4); three (luteolin, palmarin, nardostachysin) fall in the intersection.

These three consensus leads are drawn from three *different* Nepali species (Table 15), each with centuries of ethnopharmacological use against inflammatory and proliferative conditions.

Table 15: Cross-model consensus leads predicted drug-like and active by both best models, with their Nepali plant sources.

Compound	Plant (common name)	RF $\hat{p}$	GCN $\hat{p}$	QED	Chemotype
Luteolin	<i>T. chebula</i> (Haritaki)	0.679	0.514	0.511	Flavone
Palmarin	<i>T. cordifolia</i> (Giloy)	0.604	0.553	0.591	Furanoditerpenoid
Nardostachysin	<i>N. jatamansi</i> (Jatamansi)	0.584	0.592	0.526	Sesquiterpenoid ester

Luteolin (*Terminalia chebula*, Haritaki)¹ is the top consensus lead (RF $\hat{p}$ = 0.679, GCN $\hat{p}$ = 0.514; MW = 286.2 Da, QED = 0.511). It is a 3',4',5,7-tetrahydroxyflavone—a planar, hydroxylated flavone of exactly the class documented as ATP-competitive kinase binder, giving it clear biological plausibility. Crucially, luteolin is *not* present in the ChEMBL training data, so, unlike the recovered apigenin and kaempferol, it is a genuinely prospective prediction. **Palmarin** (*Tinospora cordifolia*, Giloy; RF 0.604, GCN 0.553) and **nardostachysin** (*Nardostachys jatamansi*, Jatamansi; RF 0.584, GCN 0.592) contribute a furanoditerpenoid and a sesquiterpenoid-ester scaffold respectively—chemotypes structurally distinct from luteolin and from the synthetic quinazoline inhibitors of the training set, and therefore distinct binding-mode hypotheses.

¹All compound-to-plant attributions in this section follow the organism annotations recorded in COCONUT 2.0; luteolin is a widely distributed flavone reported across many taxa, and is here associated with *Terminalia chebula* through that database rather than as a species-exclusive constituent.

That the consensus set spans three traditional species—*Terminalia chebula*, *Tinospora cordifolia*, and *Nardostachys jatamansi*, all inexpensive and locally accessible in Nepal—makes it an attractive, diversified starting point for experimental testing. **Luteolin** is the single highest-priority candidate, being both the top-ranked consensus compound and a mechanistically plausible flavone, though its moderate (< 0.80) probabilities mean it remains a computational hypothesis requiring *in vitro* confirmation. The three source species are shown in Figure 14, the corresponding compound structures in Figure 15, and their model predictions and physicochemical profiles in Figure 16.

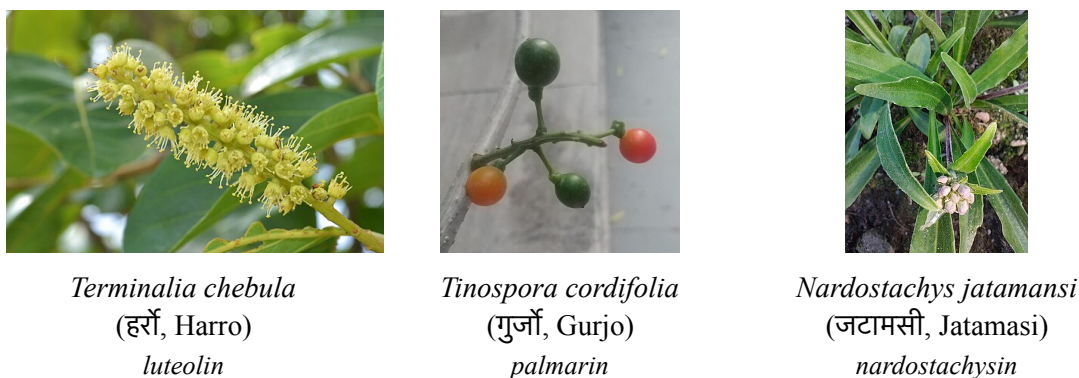


Figure 14: Consensus-lead source plants with Nepali names. Photos via Wikimedia Commons: *T. chebula* © Delonix CC BY-SA 3.0; *T. cordifolia* © Gurnavinder Sidhu CC BY-SA 4.0; *N. jatamansi* © Plantagenet CC BY-SA 4.0.

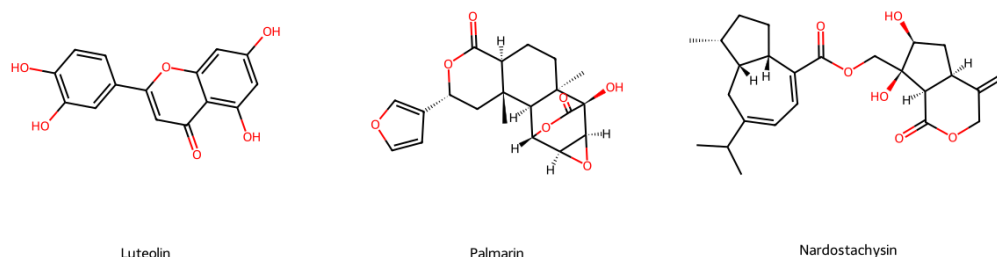


Figure 15: Chemical structures of the three consensus leads (luteolin, palmarin, nardostachysin), drawn with RDKit from COCONUT 2.0 SMILES (Sorokina et al., 2021).

4.3.5 Structural Diversity of Plant-Derived Leads

Across the two best models the genuinely unseen drug-like leads span several distinct chemotypes: xanthenes (isogentisin), the flavone luteolin, clerodane furanoditerpenoids (columbin, palmarin, tinosporaside), iridoid esters (jatamanvaltrate H, nardostachysin), and sesquiterpenoids (rulepidadiene A). This diversity suggests that several distinct binding modes to the EGFR ATP cleft may be accessible from Nepali plant compounds; luteolin—the one flavone endorsed by both best models—is the strongest single structural signal, and a richer starting point than a structurally homogeneous synthetic inhibitor library.

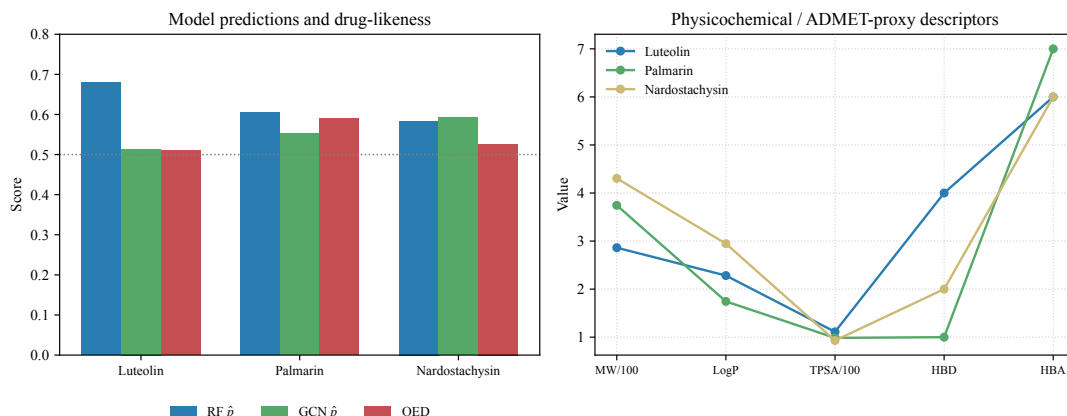


Figure 16: Model predictions and physicochemical descriptors of the three consensus leads; all satisfy Lipinski’s rule of five.

4.4 Discussion

4.4.1 Evaluation of the Research Hypotheses

The two research hypotheses formulated in Chapter 1 are evaluated against the results reported above.

H_1 (Performance Hypothesis) — not supported. H_1 predicted that the graph neural networks, with rich atom and bond featurisation and Bayesian hyperparameter optimisation, would achieve superior AUROC to the Random Forest and MLP fingerprint baselines under the Bemis-Murcko scaffold split. The results do not support this hypothesis: the fingerprint baselines achieved the highest scaffold test-set AUROC (Random Forest 0.8827, MLP 0.8547), and no GNN matched them—GCN was the strongest GNN at 0.8454, and the tuned AttentiveFP reached a validation AUROC of only 0.8632 with a test AUROC of just 0.7970. H_1 is therefore **rejected** for this dataset: on a medium-sized, scaffold-split EGFR bioactivity set, well-tuned fingerprint models remained more accurate than the graph neural networks. Although Bayesian optimisation was applied only to AttentiveFP (Section 4.2.2), this is a conservative test of H_1 rather than a confound: tuning the *weakest* default GNN still left it below every baseline, while the *strongest* default GNN (GCN, 0.8454) already trailed the Random Forest by 0.037 AUROC, so closing the gap would have required the untuned architectures to gain more from tuning than the tuned one did—an implausible margin on this dataset. This is itself an informative, honestly reported result: it directly echoes the systematic comparison of Jiang et al. (2021), who likewise found that descriptor-based models (including Random Forest) outperform graph-based models on average for molecular property prediction, and is consistent with the broader observation that ECFP4-based fingerprints are strong baselines on datasets of this scale.

H_2 (Virtual Screening Hypothesis) — not supported. H_2 predicted that, under *each* of the study’s two best models, the screening pipeline would identify at least one fully Lipinski-compliant natural-product lead with predicted activity $\hat{p} \geq 0.80$ and QED ≥ 0.40 . A preliminary screen appeared to satisfy this for the Random Forest (apigenin, $\hat{p} = 0.913$; kaempferol, $\hat{p} = 0.834$), but both compounds were found to be present in the ChEMBL training data—so their high scores are memorised known actives, not prospec-

tive predictions, and they were removed. On the resulting genuinely unseen library, *neither* best model identifies a Lipinski-compliant compound at $\hat{p} \geq 0.80$ (Random Forest and GCN both yield zero such leads); the highest prospective drug-like scores are isogentisin (RF $\hat{p} = 0.728$) and luteolin (RF 0.679, GCN 0.514). H_2 is therefore **rejected** at its stated $\hat{p} \geq 0.80$ threshold. The pipeline does still surface *moderate*-confidence, drug-like, cross-model consensus candidates (luteolin foremost) that are worth experimental follow-up, but it does not deliver the high-confidence prospective lead that H_2 posited—an honestly reported outcome once training-set leakage is properly excluded.

4.4.2 Performance of Graph Neural Networks Relative to Baselines

The results demonstrate that the relationship between GNN and traditional ML performance is nuanced. Across both the default and tuned configurations, the fingerprint-based baselines outperformed all GNN models on the scaffold test set (Random Forest AUROC 0.8827 and MLP 0.8547, versus 0.8087–0.8454 for the GNNs). This finding is consistent with the molecular machine learning literature and reflects two factors: (i) ECFP4 fingerprints encode decades of human cheminformatics knowledge through their circular neighbourhood construction, providing a strong structural inductive bias; and (ii) medium-sized bioactivity datasets ($\sim 10,000$ compounds) often favour sample-efficient ensemble methods over parameter-intensive deep learning approaches.

Bayesian hyperparameter optimisation raised AttentiveFP’s validation AUROC to 0.8632 (mid-field on the validation metric), but its tuned scaffold test AUROC (0.7970) remained the lowest of the GNNs and well below the fingerprint baselines. On this dataset, therefore, the GNNs offered no accuracy advantage: the Random Forest was the strongest classifier and, in the prospective screen, also the better-calibrated scoring model. The value of the graph approach here is not superior accuracy but a genuinely independent representation—operating directly on molecular graphs with the 29-dimensional atomic node features and 12-dimensional edge features that are invisible to ECFP4 fingerprints—which lets the best GNN (GCN) serve as a complementary second opinion to the Random Forest and thereby supports the cross-model consensus prioritisation used in the screen that follows.

4.4.3 Significance of Scaffold-Based Evaluation

A critical methodological contribution of this thesis is the strict application of Bemis-Murcko scaffold splitting. This approach fundamentally differs from random splitting, which can lead to optimistic performance estimates when structurally similar molecules appear in both training and test sets. Under scaffold splitting, the test set is composed entirely of compound-scaffold pairs whose core frameworks were never seen during training, providing a stringent evaluation of the model’s ability to generalise to genuinely novel chemical matter.

4.4.4 Ethnopharmacological Rationale for Natural Product Screening

The hit rate observed in the Nepali plant screening was itself model-dependent, ranging from 4.6% (GCN) through 34.0% (Random Forest) at $\hat{p} \geq 0.50$ on the de-duplicated library. That the models recovered the known EGFR-active flavones apigenin and

kaempferol (before these were excluded as leakage), and that the sole remaining unseen flavone—luteolin—is endorsed by both best models, is consistent with the ethnopharmacological enrichment inherent in traditional medicinal plant selection: plants used for centuries against inflammatory and proliferative conditions are biologically pre-filtered for bioactive chemical space. The planar, hydroxylated flavone scaffold can hydrogen-bond to the kinase hinge region through its ring-hydroxyl and 4-carbonyl groups—the same anchoring interaction exploited by the ATP-competitive pharmacophore of synthetic EGFR inhibitors—and flavonoids such as quercetin are long-established ATP-competitive kinase inhibitors, giving luteolin clear structural plausibility as an EGFR binder (a plausibility that nonetheless requires the *in vitro* confirmation noted above). At the same time, both the weakest model (AttentiveFP) and the best models still rank large, non-drug-like polyphenols and triterpenoids among their highest-scoring compounds—a reminder that predictions on chemical space distant from the training distribution must be interpreted with caution and coupled with drug-likeness filtering.

4.4.5 Lead Compound Prioritisation Strategy

The prioritisation strategy applied four criteria—exclusion of compounds already in the training data (to keep the screen prospective), predicted activity, Lipinski Rule of Five compliance, and consensus between the two best models—reflecting novelty, target-engagement potential, oral bioavailability, and robustness to the model-sensitivity documented above. Under this strategy **luteolin** is the top cross-model consensus lead and **isogentisin** the highest-confidence single-model prospective lead, both at moderate confidence. Critically, these compounds are derived from plants (*Terminalia chebula*, *Sweretia chirayita*) that are inexpensive, culturally familiar, and locally accessible throughout Nepal, offering a potential pathway to affordable candidate compounds without the high synthesis costs of novel synthetic scaffolds.

It is important to note that the prioritisation in this study represents an *in silico* hypothesis, not experimental confirmation. The identified leads require: (i) *in vitro* EGFR kinase inhibition assay; (ii) cellular antiproliferative testing in EGFR-mutant NSCLC cell lines (e.g., PC9, HCC827); and (iii) selectivity profiling against a kinase panel.

4.4.6 Limitations of the Current Results

Several limitations of the results presented in this chapter should be acknowledged:

- **Model-sensitivity of the screen:** The single most important limitation is that the virtual-screening hit list depends strongly on the choice of classifier (143 predicted actives for AttentiveFP versus 24 for GCN and 179 for Random Forest at $\hat{p} \geq 0.50$; 71 versus 0 versus 0 at $\hat{p} \geq 0.90$). No single model can be treated as authoritative, and the predicted probabilities are not calibrated on out-of-distribution natural products. Consensus across the best models mitigates but does not eliminate this uncertainty; only experimental assay can resolve it.
- **Train/screen overlap (leakage):** Eight of the 535 retrieved COCONUT compounds—including the apparent top hits apigenin and kaempferol—were already present in the ChEMBL training data as known actives. These were excluded so that the reported screen is genuinely prospective, but their presence

shows that natural-product libraries can overlap curated bioactivity databases, and that such overlap must be removed before any prospective claim; a screen that omits this step would badly overstate its “discoveries.”

- **Validation AUROC vs. Test AUROC:** The tuned AttentiveFP’s validation AUROC (0.8632) exceeded its scaffold test AUROC (≈ 0.80), a gap that reflects optimisation against the validation partition. Future work should employ a nested cross-validation scheme for unbiased test-set estimation.
- **Assay heterogeneity:** Training labels aggregate IC_{50} values from hundreds of laboratories with different protocols, imposing a label noise ceiling on scaffold-split test-set performance.
- **ADMET-proxy limitations:** The physicochemical properties used do not capture metabolic stability, P-glycoprotein efflux, plasma protein binding, or *in vivo* pharmacokinetics.
- **COCONUT 2.0 annotation completeness:** Organism taxonomy annotations in COCONUT 2.0 may be incomplete for some Nepali plant species, potentially missing additional bioactive compounds from these plants.
- **No 3D structural validation:** The absence of molecular docking against EGFR crystal structures means no binding pose or free energy estimate is available to corroborate model predictions for the top leads.

Chapter 5

Conclusions and Recommendations

5.1 Conclusions

This thesis presented a comprehensive, open-source end-to-end computational framework for the binary classification of EGFR inhibitor activity and the prospective virtual screening of novel lead compounds for drug-resistant NSCLC. The following principal conclusions can be drawn:

1. **Data quality and curation are foundational.** Applying rigorous ChEMBL curation—including filtering for exact IC_{50} values, pIC_{50} conversion, RDKit-based salt removal, canonicalisation, and molecular weight filtering—produced a high-quality, unambiguous dataset of 10,523 unique EGFR-bioactive compounds suitable for deep learning.
2. **Fingerprint baselines outperformed the graph neural networks.** Under strict Bemis-Murcko scaffold evaluation, the Random Forest baseline achieved the highest test AUROC (0.8827), ahead of the MLP on ECFP4 fingerprints (0.8547) and the five graph neural networks (0.8087–0.8454). Among the GNNs, GCN was strongest at default settings, while AttentiveFP—which natively exploits the 29-dimensional atom and 12-dimensional bond features—required Bayesian hyperparameter optimisation to reach mid-field on the validation set (best validation AUROC 0.8632), although its test AUROC (0.7970) remained the lowest of the GNNs. These results show that, on a medium-sized scaffold-split EGFR dataset, well-tuned fingerprint models remain more accurate than graph neural networks for classification, and hypothesis H_1 is therefore **rejected**.
3. **Nepali medicinal plant virtual screening is model-sensitive and, once leakage is removed, yields only moderate-confidence leads.** Of the 535 retrieved compounds, eight—including the apparent top hits **apigenin** and **kaempferol**—were already present in the ChEMBL training data as known actives; recovering them is a positive control, not a discovery, so they were excluded, leaving 527 genuinely unseen compounds. On this de-duplicated library the two best models gave sharply different funnels (Random Forest 179 vs. GCN 24 predicted actives at $\hat{p} \geq 0.50$), and *neither* produced a Lipinski-compliant lead at $\hat{p} \geq 0.90$ —exposing the earlier tuned-AttentiveFP screen (71 compounds above $\hat{p} \geq 0.90$) as over-confident. The strongest prospective candidates are the xanthone **isogen-tisin** (Random Forest $\hat{p} = 0.728$) and the flavone **luteolin**, the top cross-model consensus lead (both models $\hat{p} \geq 0.50$), both at moderate confidence. Because no best model reaches the $\hat{p} \geq 0.80$ bar, hypothesis H_2 is **rejected**, and the leads are consensus-supported starting points for experimental validation rather than high-confidence hits.

4. **Consensus across models is essential for trustworthy natural-product screening.** Because the predicted hit list depends strongly on the choice of classifier and the models are not calibrated on out-of-distribution natural products, no single model can be treated as authoritative. Prioritising compounds endorsed by more than one strong model—and coupling activity prediction with drug-likeness filtering, since the weaker models still ranked large, non-drug-like *Terminalia chebula* tannins and *Tinospora* triterpenoids among their highest scorers—is the key methodological lesson of the screening campaign. The flavone leads that survive this scrutiny, combined with the low cost and cultural accessibility of their source plants in Nepal, make them valuable starting points for affordable NSCLC drug discovery.
5. **The pipeline is reproducible, scalable, and regionally relevant.** The entire framework uses exclusively open-source Python tools (PyTorch Geometric, RDKit, Optuna, scikit-learn, COCONUT 2.0) and public databases (ChEMBL, COCONUT), runs on commodity CPU hardware, and is immediately adaptable to other kinase targets or plant species without specialised hardware or licensed software. This makes it directly accessible to academic researchers in Nepal and other resource-limited South Asian settings.

5.2 Recommendations for Future Work

1. **Experimental validation of Nepali plant leads.** The top genuinely unseen, cross-model-consensus leads—particularly the flavone luteolin (*Terminalia chebula*) and the xanthone isogentisin (*Swertia chirayita*)—should be subjected to *in vitro* EGFR kinase inhibition assays and EGFR-mutant NSCLC cell line proliferation assays (e.g., PC9, HCC827) to validate the computational predictions empirically. The known actives recovered by the screen (apigenin, kaempferol) can serve as assay positive controls.
2. **Mutant-specific modelling.** As mutation-specific bioactivity datasets for T790M and C797S EGFR grow in public databases, dedicated GNN models trained specifically on these mutant targets should be developed to directly address drug resistance.
3. **3D graph representation.** Incorporating three-dimensional molecular conformations (e.g., through SchNet or DimeNet++ architectures) and protein structure-aware approaches could improve the prediction of binding affinity for specific EGFR mutant forms.
4. **Pre-training on large molecular datasets.** Applying self-supervised graph pre-training on millions of unlabelled molecules (e.g., from ZINC or PubChem) before fine-tuning on EGFR data could further improve scaffold generalisation.
5. **Broader Nepali plant species survey.** The current study covered seven ethnopharmacologically relevant species. A systematic expansion to the full catalogue of 700+ documented Nepali medicinal plants, using COCONUT 2.0 and complementary databases (e.g., Dr. Duke’s Phytochemical Database, NPACT), could identify additional EGFR inhibitor candidates from the rich Himalayan biodiversity.

6. **Extended ADMET profiling.** The current ADMET analysis used RDKit physicochemical property proxies. Future work should incorporate dedicated ADMET prediction tools such as SwissADME, pkCSM, or deep ADMET models for comprehensive absorption, distribution, metabolism, excretion, and toxicity profiling of the top leads.
7. **Collaboration with Nepal-based research institutions.** To translate these computational predictions into experimental evidence, partnerships should be established with Nepal-based pharmacology and oncology research institutions—such as the B.P. Koirala Institute of Health Sciences (BPKIHS) and the National Cancer Hospital and Research Centre (NCHRC)—for wet-lab assay, phytochemical isolation, and clinical-translational validation of the top Nepali medicinal plant lead compounds.

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Appendix A: Dataset Information

A.1 ChEMBL Query Parameters

The EGFR bioactivity dataset was retrieved from ChEMBL (Mendez et al., 2019) via the REST API in May 2026 using the parameters listed in Table 16. Only binding assay records with high confidence scores were retained.

Table 16: ChEMBL query parameters used to retrieve EGFR bioactivity data.

Parameter	Value
Database	ChEMBL (May 2026 release)
Target name	Epidermal Growth Factor Receptor (EGFR)
ChEMBL target ID	CHEMBL203
Activity type	IC ₅₀
Units	nM (nanomolar)
Assay type	B (binding assay)
Confidence score	≥ 9 (direct single protein target)
Organism	<i>Homo sapiens</i>
Activity threshold	pIC ₅₀ $\geq 6.0 \rightarrow$ Active (IC ₅₀ $\leq 1 \mu\text{M}$); pIC ₅₀ $< 6.0 \rightarrow$ Inactive
Data split method	Bemis-Murcko scaffold split

A.2 Dataset Curation Statistics

Table 17 shows the number of compounds retained at each stage of the curation pipeline.

Table 17: Step-by-step dataset curation statistics for the EGFR bioactivity dataset.

Step	Filter Applied	Compounds Remaining
0	Raw ChEMBL records retrieved	~15,000–20,000
1	Remove missing or invalid SMILES	—
2	Salt stripping and canonicalisation	—
3	Molecular weight filter (100–1000 Da)	—
4	Remove duplicate molecules	—
Final curated dataset		10,523
Active (pIC ₅₀ ≥ 6.0)		7,792 (74.0%)
Inactive (pIC ₅₀ < 6.0)		2,731 (26.0%)
Active:Inactive ratio		2.85:1
Training set (80%, scaffold)		8,418
Validation set (10%, scaffold)		1,052
Test set (10%, scaffold)		1,053

The Bemis-Murcko scaffold split ensures that all molecules sharing the same core framework are assigned to the same partition, preventing optimistic bias from scaffold mem-

orisation and providing a more realistic estimate of generalisation to novel chemical matter.

Appendix B: Screening Library and Virtual Screening Results

B.1 Screening Library Source

The virtual screening library was curated from seven traditional Nepali medicinal plant species—*Swertia chirayita*, *Rhododendron arboreum*, *Nardostachys jatamansi*, *Terminalia chebula*, *Berberis aristata*, *Ocimum sanctum*, and *Tinospora cordifolia*—retrieved from the COCONUT 2.0 natural product database (Sorokina et al., 2021) and standardised with the same RDKit pipeline as the training data. Retrieval returned 535 compounds; **eight already present in the ChEMBL data** (including apigenin and kaempferol) were removed as train/screen leakage, leaving **527 genuinely unseen compounds** screened. These ethnomedicinal species are traditionally used for inflammatory and proliferative conditions, enriching the library for bioactive chemical space relative to random chemical libraries. The composition of the screened library by species is given in Table 18.

Table 18: Composition of the screened Nepali plant library by species (COCONUT 2.0).

Plant Species	Major Compound Class	Compounds
<i>Nardostachys jatamansi</i>	Sesquiterpenes, coumarins	199
<i>Terminalia chebula</i>	Hydrolysable tannins	192
<i>Tinospora cordifolia</i>	Diterpenes, alkaloids	71
<i>Swertia chirayita</i>	Xanthones, secoiridoids	23
<i>Berberis aristata</i>	Isoquinoline alkaloids	19
<i>Rhododendron arboreum</i>	Triterpenoids, flavonoids	15
<i>Ocimum sanctum</i>	Phenylpropanoids, terpenoids	8
Total	—	527

All 527 unseen compounds were scored by the study’s two best-performing models—the best graph network (GCN) and the best overall classifier (Random Forest). The best classifier predicted **179 compounds** active at $\hat{p} \geq 0.50$; on the de-duplicated library *neither* best model produced a Lipinski-compliant lead above $\hat{p} \geq 0.80$. The highest-confidence unseen drug-like lead is **isogentisin** (Random Forest $\hat{p} = 0.728$), and **luteolin** is the top cross-model consensus lead (predicted active by both models).

B.2 Top Prospective Lipinski-Compliant Lead

Table 19: Canonical SMILES and physicochemical properties of the highest-confidence *genuinely unseen* Lipinski-compliant lead (Random Forest).

Property	Value
Compound	Isogentisin
Source species	<i>Swertia chirayita</i>
Predicted probability ($\hat{p}$, Random Forest)	0.7282
Predicted probability ($\hat{p}$, GCN)	0.4600
Canonical SMILES	<chem>C0c1ccc2oc3cc(0)cc(0)c3c(=O)c2c1</chem>
Molecular Weight (Da)	258.2
LogP	2.37
QED	0.655
TPSA ($\text{\AA}^2$)	79.9
Lipinski Ro5 violations	0
Compound class	Xanthone (polyphenolic)

B.3 Virtual Screening Summary Statistics

Table 20: Virtual screening summary for the two best models on the 527-compound Nepali library.

Metric	Random Forest	GCN
Total compounds screened	527	527
Predicted active ($\hat{p} \geq 0.50$)	179 (34.0%)	24 (4.6%)
Predicted active ($\hat{p} \geq 0.75$)	32 (6.1%)	2 (0.4%)
Predicted active ($\hat{p} \geq 0.90$)	0 (0.0%)	0 (0.0%)
Highest-confidence compliant lead	Isogentisin	Rulepidadiene A
Top consensus lead (both models)	Luteolin	

Appendix C: GNN Architecture and Hyperparameter Optimisation

C.1 AttentiveFP Final Hyperparameters

Table 21 lists the final hyperparameters of the tuned AttentiveFP model selected by Bayesian optimisation using Optuna over 20 trials, with validation AUROC as the objective.

Table 21: Final hyperparameters of the tuned AttentiveFP model selected by Optuna Bayesian optimisation.

Hyperparameter	Search Range	Optimal Value
Hidden dimension	{64, 128, 256}	128
Message passing layers	2–4	3
Attention timesteps (T)	2–4	2
Dropout rate	0.1–0.5	0.129
Learning rate	10^{-4} – 10^{-2}	2.61×10^{-3}
Batch size	{32, 64, 128}	64
L2 weight decay	10^{-5} – 10^{-3}	1.10×10^{-5}
Optimiser	—	Adam
Loss function	—	Binary cross-entropy
Training epochs	—	≤ 80 (early stopping)

C.2 Model Performance Comparison

Table 22 compares all five GNN architectures and two traditional machine learning baselines evaluated under the Bemis-Murcko scaffold split.

Table 22: Performance comparison of GNN architectures and traditional ML baselines on the EGFR scaffold test set (Bemis-Murcko split).

Model	AUROC	AUPRC	F1	MCC
Random Forest (ECFP4)	0.8827	0.9496	0.8762	0.5072
MLP (ECFP4)	0.8547	0.9354	0.8629	0.5281
GCN	0.8454	0.9332	0.7877	0.4707
GraphSAGE	0.8229	0.9179	0.8070	0.4706
GIN	0.8217	0.9163	0.8487	0.4865
GAT	0.8162	0.9176	0.7689	0.4473
AttentiveFP	0.8087	0.9108	0.8254	0.3989

C.3 Molecular Graph Feature Vectors

Table 23: Node (atom) and edge (bond) feature vectors used as input to all GNN architectures.

Feature	Dim.	Description
<i>Node features ($d_n = 29$)</i>		
Atom type	10	One-hot: C, N, O, S, F, Cl, Br, I, P, other
Degree	6	One-hot: 0–5 bonds
Formal charge	1	Integer scalar
Hydrogen count	5	One-hot: 0–4 H atoms
Hybridisation	5	One-hot: sp, sp2, sp3, sp3d, sp3d2
Aromaticity	1	Binary flag
Ring membership	1	Binary flag
<i>Edge features ($d_e = 12$)</i>		
Bond type	4	One-hot: single, double, triple, aromatic
Conjugation	1	Binary flag
Ring membership	1	Binary flag
Stereo	6	One-hot: none, any, E, Z, cis, trans

Appendix D: Software and Computational Environment

D.1 Software List

Table 24: Software packages, libraries, and databases used in this study.

Software / Tool	Version	Role
Python	3.14	Pipeline scripting
PyTorch	2.12.0	Deep learning framework
PyTorch Geometric	2.7.0	GNN implementation
RDKit	2026.3.2	Cheminformatics, molecular graphs
Optuna	4.8.0	Bayesian hyperparameter optimisation
scikit-learn	1.8.0	Baseline ML, metrics
pandas	3.0.3	Data handling
numpy	2.4.6	Numerical computation
matplotlib	3.10.9	Visualisation
seaborn	0.13.2	Statistical visualisation
<i>Databases</i>		
ChEMBL	May 2026	EGFR bioactivity training data
COCONUT	2.0	Nepali medicinal plant screening library
PubChem	—	Compound verification

D.2 Hardware Environment

Table 25: Computational hardware and software environment used for all experiments.

Component	Specification
Processor	Apple Silicon (M-series)
RAM	16 GB unified memory
GPU	Not used (CPU-only execution)
Operating system	macOS
Python environment	Python 3.14 virtual environment (venv)
GNN training backend	PyTorch (CPU)

All GNN training and virtual screening inference were executed entirely on CPU, without requiring a GPU, demonstrating the accessibility of the pipeline on commodity academic hardware. The full training pipeline for AttentiveFP (20 Optuna trials + final training) completed within a practical timeframe on the CPU.